PageWriter Trim I, II, III

GETTING STARTED GUIDE

About This Edition

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Warranty

Philips Medial Systems reserves the right to make changes to both this *Getting Started Guide* and to the product that it describes. Product specifications are subject to change without notice.

Nothing contained within this *Getting Started Guide* is intended as any offer, warranty, promise, or contractual condition, and must not be taken as such.

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Compliance

The Philips Medical Systems PageWriter Trim cardiograph complies with all relevant international and national standards and laws. Information on compliance will be supplied on request by a local Philips Medical Systems representative, or by the manufacturer.

Intended Use of this Getting Started Guide

This Philips product is intended to be operated only in accordance with the safety procedures and operating instructions provided in this *Getting Started Guide*, and in accordance with the purposes for which it was designed. Installation, use, and operation of this product is subject to the laws in effect in the jurisdiction(s) in which the product is being used. Users must only install, use, and operate this product in such a manner that does not conflict with applicable laws or regulations that have the force of law.Use of this product for purposes other than the express intended purpose provided by the manufacturer, or incorrect use and operation, may relieve the manufacturer (or agent) from all or some responsibility for resultant non-compliance, damage, or injury.

United States federal law restricts this device to use by or on the order of a physician. THIS PRODUCT IS NOT INTENDED FOR HOME USE.

Training

Users of this product must receive adequate clinical training on its safe and effective use before attempting to operate the product as described in this *Getting Started Guide*.

Training requirements vary by country. Users must ensure that they receive adequate clinical training in accordance with local laws or regulations.

For further information on available training on the use of this product, please contact a Philips Medical Systems representative, or the manufacturer.

Medical Device Directive

The PageWriter Trim Cardiograph complies with the requirements of the Medical Device Directive 93/42/EEC and carries the CE0123 mark accordingly.

Authorized EU-representative:

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Contents

Safety Summary

Getting Started

PageWriter Trim II and III Configuration

Suppressed Borderline Interpretive Statements

About the PageWriter Trim Cardiograph

About the Getting Started Guide

This *PageWriter Trim Cardiograph Getting Started Guide* is intended to assist users in the safe and effective use of the product.

Before attempting to operate this product, read this *Getting Started Guide,* and note and strictly observe all Warning and Cautions as described in this document.

Pay special attention to all of the safety information provided in the Safety Summary section. For more information, see page i-ii.

WARNING **Warning statements describe conditions or actions that may result in a potentially serious outcome, adverse event, or a safety hazard. Failure to follow a Warning may result in death or serious injury to the user or to the patient.**

CAUTION Caution statements describe when special care is necessary for the safe and effective use of the product. Failure to follow a caution may result in minor to moderate personal injury or damage to the product or other property, a remote risk of more serious injury, or may cause environmental pollution.

NOTE Notes contain additional important information about a topic.

TIP A Tip contains suggested information on using a particular feature.

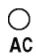
Menu items and button names appear in bold no-serif font. Example: Touch the **Config** button.

Safety Summary

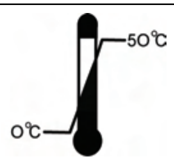
Safety and Regulatory Symbols Marked on the Cardiograph

Symbol	Name	Description
	Attention	See the *PageWriter Trim Getting Started Guide* for information.
	Type CF	ECG physio isolation is type CF, defibrillator proof. Electrical leakage current is suitable for all patient applications including direct cardiac application.
	On/Standby or On/Off	Pressing the button with this symbol on it turns on the cardiograph, turns off the cardiograph, or puts the cardiograph into Standby (power saving mode).
	Electrostatic Discharge	Do not touch exposed pins. Touching exposed pins can cause electrostatic discharge that can damage the cardiograph.
	Equipotential grounding post	Equipotential grounding post used for establishing common ground between instruments.
	Fuse	Cardiograph contains a 1.5 amp (250V) time-delay fuse.
	Input	The connector near this symbol receives an incoming signal.

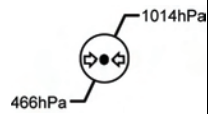
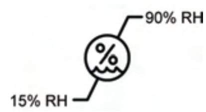
Safety and Regulatory Symbols Marked on the Cardiograph

SN	Serial Number	The number next to this symbol is the serial number of the cardiograph.
REF	Product model number	The number next to this symbol is the product model number of the cardiograph.
IPX4	Entry of liquids	The PIM (Patient Interface Module) is protected against splashing water. Water splashed against the PIM from any direction shall have no harmful effect.
	AC power indicator light	When lit, indicates that AC power is on. The battery is charging when inserted into the cardiograph.
	Disposal Information	Dispose of in accordance with the requirements of your country.

Safety Symbols Marked on the Cardiograph Packaging

Symbol	Description
	Keep dry.
	Ambient temperature range of 0 °C (32°·F) to 50 °C (122° F) (non-condensing) for transport and storage.

Safety Symbols Marked on the Cardiograph Packaging

	Atmospheric pressure range of 466 hPa to 1014 hPa for transport and storage.
	Relative humidity range of 15% to 90% (non-condensing) for transport and storage.
	Move and store packaging this end up.
	Fragile.
	Sealed lead acid battery. Do not dispose of in trash. Follow local regulations for disposing of as small chemical waste.

Important Patient and Safety Information

The PageWriter Trim cardiograph isolates all connections to the patient from electrical ground and all other conductive circuits in the cardiograph. This reduces the possibility of hazardous currents passing from the cardiograph through the patient's heart to ground, and from other equipment connected to the patient passing through the leads into the cardiograph to ground.

WARNING **Failure to follow these warnings could affect both patient and operator safety.**

- When operating the cardiograph on AC power, ensure that the cardiograph and all other electrical equipment connected to or near the patient are effectively grounded.
- Use only grounded power cords (three-wire power cords with grounded plugs) and grounded electrical outlets. **Never** adapt a grounded plug to fit an ungrounded outlet by removing the ground prong. Use the equipotential post when redundant earth ground is necessary according to IEC 60601-1-1.
- If a safe ground connection is not ensured, operate the cardiograph on battery power only.
- The use of equipment that applies high frequency voltages to the patient (including electrosurgical equipment and some respiration transducers) is not supported and may produce undesired results. Disconnect the patient data cable from the cardiograph, or detach the leads from the patient prior to performing any procedure that uses high frequency surgical equipment.
- Do not perform ST analysis on the R/T ECG screen display or on Rhythm reports when the 0.5 Hz Baseline Wander filter is applied.
- If abnormal ECG data appears on the printed report, and the abnormal data does not have a physiological origin, perform the printer diagnostic test to assess printer performance.
- When printing a Rhythm report, there may be a slight delay before the Rhythm report begins to print on the cardiograph. Rhythm printing is not completed in real-time.
- Pace pulse tick marks will not print on an Auto ECG that uses simultaneous acquisition.

WARNING **Do not touch accessible connector pins and the patient simultaneously.**

Electrical shock hazard. Keep cardiograph, Patient Interface Module (PIM) and all cardiograph accessories away from liquids. Do not immerse cardiograph, PIM, or other accessories in any liquids.

- Periodically inspect the patient data cable, lead wires, and AC power cord for any worn or cracked insulation to ensure that no inner conductive material is exposed. Discard worn accessories and replace them only with Philips Medical Systems accessories, see page 1-46 for more information.
- Keep the patient data cable away from power cords and any other electrical equipment. Failure to do so can result in AC power line frequency interference on the ECG trace.
- The Philips Medical Systems patient data cable (supplied with cardiograph) is an integral part of the cardiograph safety features. Use of any other patient data cable may compromise defibrillation protection, degrade cardiograph performance, and may result in distorted ECG data.
- Only qualified personnel may service the cardiograph or may open the cardiograph housing to access internal cardiograph components. Do not open any covers on the cardiograph. There are no internal cardiograph components that are serviced by the operator.
- Do not use this cardiograph near flammable anesthetics. It is not intended for use in explosive environments or in operating rooms.
- Do not touch the patient, the patient data cable, any unused patient leads, or the cardiograph during defibrillation. Death or injury may occur from the electrical shock delivered by the defibrillator.
- Always use electrode gel with reusable electrodes during defibrillation as ECG recovery will be greater than 10 seconds. Philips Medical Systems recommends the use of disposable electrodes at all times.
- Ensure that the electrodes or lead wires do not come in contact with any other conductive materials (including earth-grounded materials) especially when connecting or disconnecting electrodes to or from a patient.
- Connecting multiple medical electrical equipment to the same patient may pose a safety hazard due to the summation of leakage currents. Any combination of instruments should be evaluated by local safety personnel before being put into service.

- Portable medical equipment such as X-rays and MRI may produce electromagnetic interference that produces noise in the ECG signal. Move the cardiograph away from these potential sources of electromagnetic interference.
- Do not pull on the paper while an ECG report is being printed. This can cause distortion of the waveform and can lead to potential misdiagnosis.
- Only use the Philips Medical Systems AC power cord supplied with the cardiograph. Periodically inspect the AC power cord and AC power connector (rear of cardiograph, see page 1-10) to ensure that both are in a safe and operable condition. If the AC power cord or AC power connector is not in a safe or operable condition, operate the cardiograph on battery power and contact Philips Medical Systems for service.
- The cardiograph has been safety tested with the recommended accessories, peripherals, and leads, and no hazard was found when the cardiograph is operated with cardiac pacemakers or other stimulators.
- Do not connect any equipment or accessories to the cardiograph that are not manufactured or approved by Philips Medical Systems or that are not IEC 60601-1 approved. The operation or use of non-approved equipment or accessories with the cardiograph is not tested or supported, and cardiograph operation and safety are not guaranteed.
- The list of cables and other accessories with which Philips claims compliance with the emissions and immunity requirements of IEC standard 60601-1-2 are listed in “Supplies and Ordering Information” on page 1-46.

WARNING **When using additional peripheral equipment powered from an electrical source other than the cardiograph, the combination is considered to be a medical system. It is the responsibility of the operator to comply with IEC 60601-1-1 and test the medical system according to the requirements. For additional information contact Philips Medical Systems.**

WARNING **Do not use non-medical peripherals within 1.83 meters or 6 feet of a patient unless the non-medical peripherals receive power from the cardiograph or from an isolation transformer that meets medical safety standards.**

- Only install Philips Medical Systems software on the cardiograph. The installation or use of software not approved by Philips Medical Systems is strictly prohibited and cardiograph safety and performance are not guaranteed.
- Only use Philips Medical Systems replacement parts and supplies with the cardiograph. The use of non-approved replacement parts and supplies with the cardiograph is strictly prohibited. Cardiograph safety and performance are not guaranteed when non-approved replacement parts and supplies are used with the cardiograph.
- Manual measurements of ECG intervals and magnitudes should be performed on printed ECG reports only. Do not make manual measurements of ECG intervals and magnitudes on the R/T ECG display since these ECG representations are scaled.
- Only use patient electrodes that are approved by Philips Medical Systems. The use of non-approved patient electrodes may degrade cardiograph performance.
- The Philips Medical Systems warranty is applicable only if you use Philips Medical Systems approved accessories and replacement parts. See "Supplies and Ordering Information" on page 1-46 for more information.
- Before using the Patient Cable Arm with the cardiograph cart, properly install the counter weight on the cardiograph base.
- Only use the shielded LAN cable provided with the PageWriter Trim cardiograph, Philips Part Number 989803138021. Do not use any other LAN cables with the PageWriter Trim cardiograph. Use of unapproved LAN cables may result in radiated emissions that exceed the limit specified by CISPR11 Class B.

- The combined maximum weight that can be placed on the cardiograph cart shelf and the top surface of the cart cannot exceed 20 kg (44 lbs). Do not place more than the specified weight on the cardiograph top surface and shelf.
- Do not connect any device to the RS-232 port on the rear of the cardiograph when the patient data cable is connected to a patient.
- There are no cardiograph parts that can be sterilized.
- The cardiograph is not intended for direct, or invasive cardiac monitoring purposes.
- Excessive, repetitive use of the cardiograph keyboard and the cardiograph Trim Knob may result in a risk of developing carpal tunnel syndrome.
- Ensure that the patient data cable is tucked away from the cardiograph cart wheels when transporting the cardiograph. Ensure that the patient data cable does not present a hazard when pushing the cardiograph cart.
- For information on the standard IEC 60601-2-52, please go to the Philips InCenter web site (incenter.medical.philips.com). For information on using the Philisp InCenter site, see page 1-4.

The PageWriter Trim Cardiograph

This Philips product is intended to be used and operated only in accordance with the safety procedures and operating instructions provided in this *Getting Started Guide*, and for the purposes for which it was designed. The purposes for which the product is intended are provided below. However, nothing stated in this *Getting Started Guide* reduces the responsibility of the user for sound clinical judgment and best clinical procedures.

Intended Use

The intended use of the cardiograph is to acquire multi-channel ECG signals from adult and pediatric patients from body surface ECG electrodes and to record, display, analyze, and store these ECG signals for review by the user. The cardiograph is to be used in healthcare facilities by trained healthcare professionals. Analysis of the ECG signals is accomplished with algorithms that provide measurements, data presentations, graphical presentations, and interpretations for review by the user.

The interpreted ECG with measurements and interpretive statements is offered to the clinician on an advisory basis only. It is to be used in conjunction with the clinician's knowledge of the patient, the results of the physical examination, the ECG tracings, and other clinical findings. A qualified physician is asked to overread and validate (or change) the computer-generated ECG interpretation.

Indications for Use

The cardiograph is to be used where the clinician decides to evaluate the electrocardiogram of adult and pediatric patients as part of decisions regarding possible diagnosis, potential treatment, effectiveness of treatment, or to rule out causes for symptoms.

The Philips 12-Lead Algorithm

The PageWriter Trim cardiograph software uses the Philips 12-Lead Algorithm. The algorithm in the software analyzes the morphology and rhythm on each of the 12 leads and summarizes the results. The set of summarized measurements is then analyzed by the clinically-proven ECG Analysis Program.

12-lead Reports may include or exclude ECG measurements, reasons, or analysis statements.

Intended Use

The intended use of the Philips 12-Lead Algorithm is to analyze multi-channel ECG signals from adult and pediatric patients with algorithms that provide measurements, data presentations, graphical presentations, and interpretations for review by the user.

The interpreted ECG with measurements and interpretive statements is offered to the clinician on an advisory basis only. It is to be used in conjunction with the clinician's knowledge of the patient, the results of the physical examination, the ECG tracings, and other clinical findings. A qualified physician is asked to overread and validate (or change) the computer-generated ECG interpretation.

Indications for Use

The Philips 12-Lead Algorithm is to be used where the clinician decides to evaluate the electrocardiogram of adult and pediatric patients as part of decisions regarding possible diagnosis, potential treatment, effectiveness of treatment, or to rule out causes for symptoms.

1

Getting Started

Welcome to the PageWriter Trim cardiograph! With its fast and simple operation, clearly labeled PIM and lead wires, and convenient cart, it is the ideal cardiograph for processing large volumes of ECGs quickly and easily.

The PageWriter Trim supports multiple ECG acquisition modes, display settings, and report formats. Signal quality indicators provide instant feedback to the user to help ensure a quality ECG each and every time.

This *PageWriter Trim Cardiograph Getting Started Guide* includes an overview of the cardiograph and all of its components, setup and assembly instructions, and information on ordering supplies and accessories. It is intended to be used with the other materials included in the PageWriter Trim Cardiograph Learning Kit.

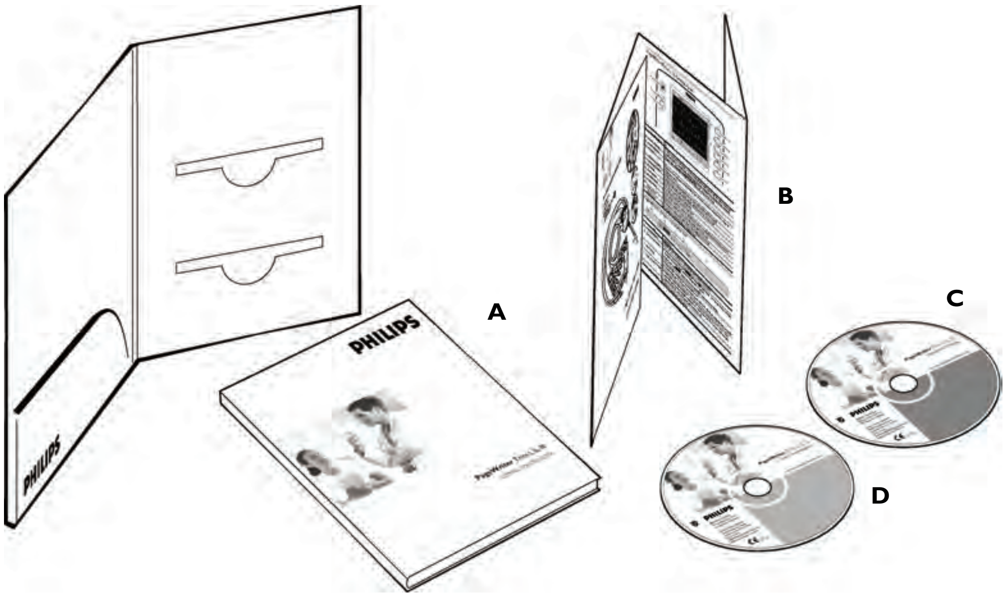
NOTE Read and complete the materials included in the PageWriter Trim Cardiograph Learning Kit before using the cardiograph. Pay close attention to all warnings and cautions.

PageWriter Trim Cardiograph Learning Kit

Philips Medical Systems provides detailed instructional and reference materials in the PageWriter Trim Learning Kit

The PageWriter Trim Learning Kit contains the *Getting Started Guide*, a Quick Help Card, and the User Documentation and Interactive Training CDs.

Figure 1-1 The PageWriter Trim Cardiograph Learning Kit



About the PageWriter Trim Learning Kit

- *Getting Started Guide* (**A**)

  This guide includes important information that must be read before operating the cardiograph. It includes an overview of cardiograph features and functions, assembly and setup instructions, how to configure the cardiograph, and how to order supplies.

- *Quick Help Card* (**B**)

  This card provides quick reference information about basic cardiograph features, lead placement, and signal quality indicators. It is also provided as a PDF file on the PageWriter Trim User Documentation CD.

- *PageWriter Trim User Documentation CD* (**C**)

  The User Documentation CD includes the following files:

  – *Philips 12-Lead Algorithm Physician's Guide*

  This guide describes the Philips 12-Lead Algorithm, and lists all of the interpretive statements included in the 0A criteria.

  – *PageWriter Trim I, II, III Instructions for Use*

  This instructions for use includes complete information on all aspects of using and maintaining the cardiograph.

  – *PageWriter Trim I, II, III Quick Help Card*

  A PDF version of the fold-out *Quick Help Card* provided in the Learning Kit.

  – *PageWriter Trim Cart Assembly Instructions*

  This instruction sheet describes how to assemble the optional PageWriter Trim cart.

  – *PageWriter Trim Wireless LAN Installation Instructions*

  This instruction sheet describes how to install and configure the wireless LAN option for the cardiograph.

To view user documentation:

1. Insert the CD into a PC CD-ROM drive. The CD only works on PCs with a Windows operating system.

   The user documentation CD main menu opens automatically. Click on a blue button or the file name to view the file.

2. If the menu does not automatically appear, open the CD in Windows Explorer.

3. Double-click the file **menu.pdf**. The main menu appears. Any of the files on the CD may be printed or saved to a PC hard drive.

NOTE If files from the CD are saved to a PC hard drive, Acrobat Reader 7.0 will need to be installed on the PC in order to view the files. For a free install go to: www.adobe.com.

- *PageWriter Trim Cardiograph Interactive Training Program* CD (**D**)

  The interactive training CD contains a computer-based training program that describes how to properly record an ECG and how to operate the cardiograph.

  Use this program to train the entire ECG staff in the proper operation of the cardiograph.

To run the interactive training program:

1. Insert the CD into a CD-ROM drive. The CD only works on PCs with a Windows operating system.
   The interactive training program starts automatically.
2. If the interactive training program does not automatically start, open the CD in Windows Explorer.
3. Double-click the file **pwtrim.exe**. The interactive training program file may be saved to any PC hard drive and does not require any special software to operate.

Philips 12-Lead ECG XML Information and Tools

The PageWriter Trim cardiograph exports 12-lead ECG data in XML (Extensible Markup Language) format. The XML schema for the Philips 12-lead ECG files, and a complementary suite of XML utilities and tools are available for download from the Philips InCenter web site. An *XML Utility Suite Instructions for Use* is also available for download. This *Instructions for Use* describes how to install and configure the XML utilities. Check the InCenter site regularly for documentation updates, and for further information and updates to the XML Utility Suite.

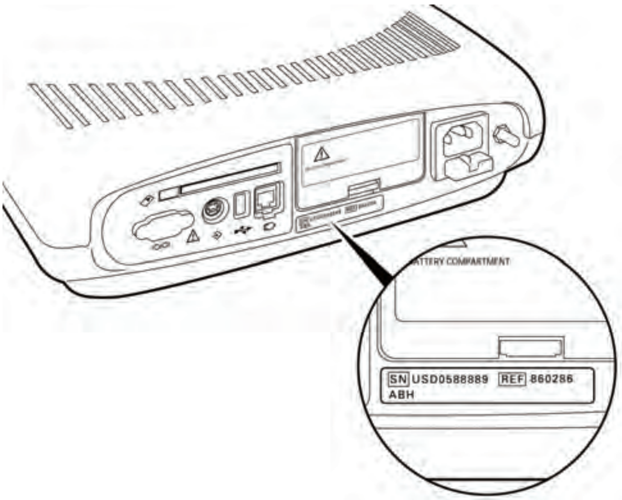
Using the Philips InCenter Site

The Philips InCenter web site provides frequent updates to all Philips Cardiac Systems product documentation and product software, including the PageWriter Trim cardiographs.

The Philips InCenter web site requires an active registration and password. To register, go to the InCenter web site at: incenter.medical.philips.com and click on the **Click here for access to software updates and documentation for cardiology products** link located on the right side of the page. The Cardiac Systems InCenter Registration page appears. Complete all of the information fields on the page to receive a login and password for the InCenter site.

Registration for the InCenter web site requires the serial number(s) for all PageWriter Trim cardiographs in active use at your facility. The serial number is located on the rear panel of the cardiograph (directly underneath the battery door), next to the **SN** text.

Figure 1-2 Serial Number on Product Identification Label



About Adobe Acrobat Versions

Adobe Acrobat Reader version 7.0 must be installed on the PC that is used to access the Philips InCenter web site. Previous versions of Acrobat Reader are not compatible, and attempting to access InCenter with a previous version of Acrobat Reader will result in error messages when opening documents. Uninstall all previous versions of Acrobat Reader, and then proceed for a free install of Acrobat Reader 7.0 at: www.adobe.com.

Adobe Acrobat Professional version 7.0 is also not compatible with the Philips InCenter site, and error messages will appear when opening documents with this version of Acrobat software. Acrobat Reader 7.0 must be installed in addition to Acrobat Professional.

Follow the procedure below when using the Philips InCenter site.

To access documents on the InCenter site:

1. Exit Acrobat Professional (if open).
2. Start Acrobat Reader 7.0.
3. Open Internet Explorer, and go to the Philips InCenter site. Keep Acrobat Reader 7.0 open the entire time while accessing the InCenter site.

Attaching the Cardiograph to the Cart

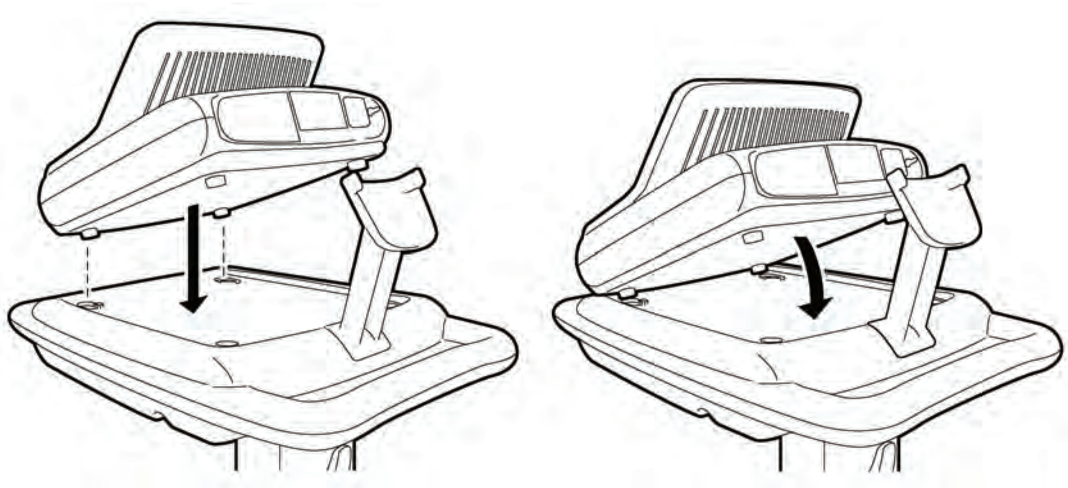
The PageWriter Trim cardiograph is available with an optional cart that includes a storage shelf and a holder for the PIM (Patient Interface Module).

For information on assembling the PageWriter Trim cart, see the *PageWriter Trim Cart Assembly Instructions* available as a PDF file on the User Documentation CD. This file may also be downloaded from the Philips InCenter web site (incenter.medical.philips.com).

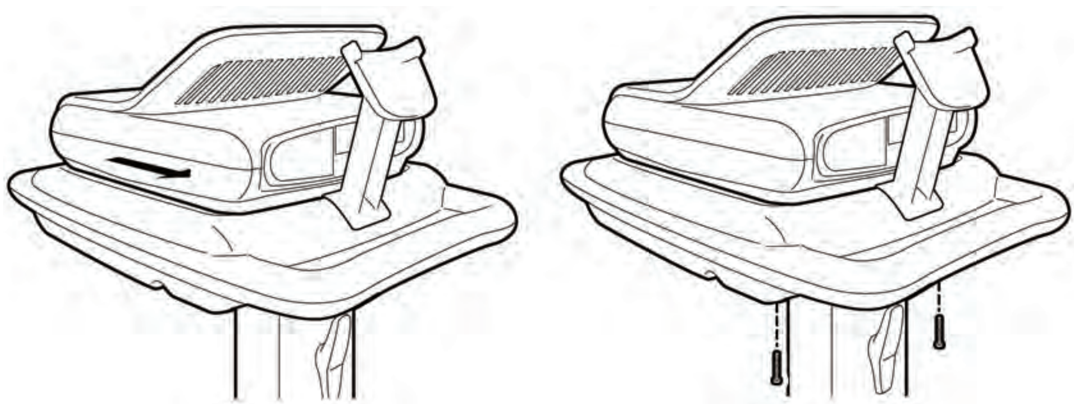
CAUTION Follow the procedure below to ensure that the cardiograph is securely fastened to the cart before use.

To attach the cardiograph to the cart:

1 Align the front feet of the cardiograph with the front locking holes on the cart. Align the rear feet of the cardiograph with the rear screw holes on the cart. Lower the cardiograph onto the cart.



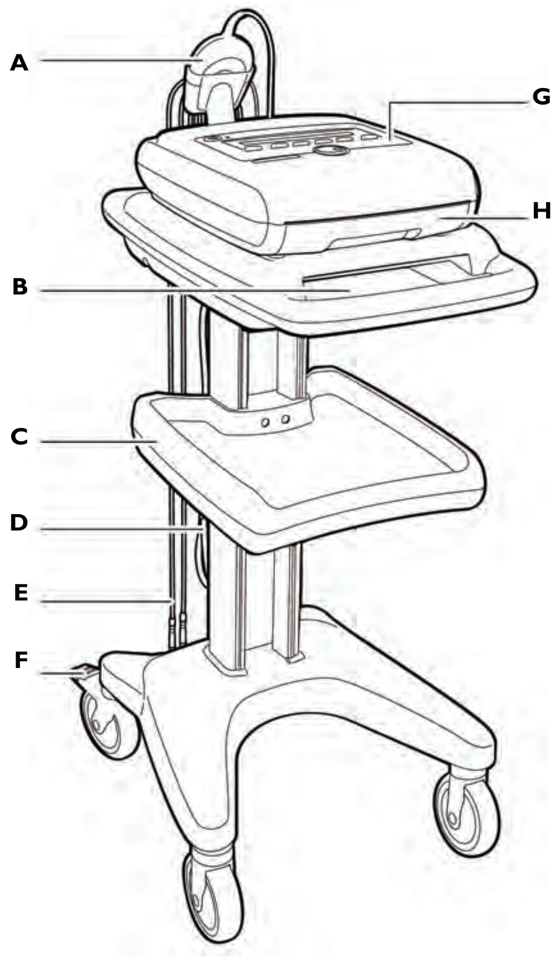
2 Slide the cardiograph back to lock the front and rear feet into place. Insert the screws through the bottom of the cart and through the screw holes. Tighten the screws with a Phillips head screwdriver.



PageWriter Trim I Cardiograph Parts

The following sections show front, side, and rear views of the PageWriter Trim I cardiograph.

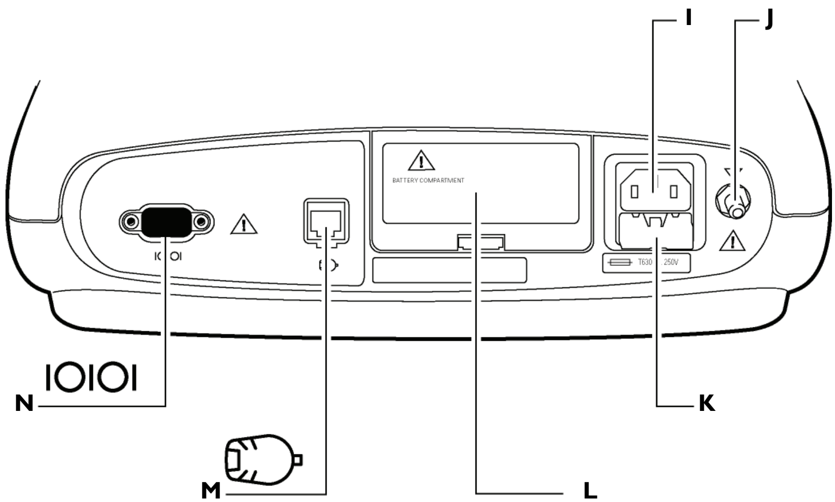
Figure 1-3 PageWriter Trim I Cardiograph and Cart (Front View)



A	Patient Interface Module (PIM)	**E**	PIM leads
B	Printer paper/report storage slot	**F**	Wheel brake
C	Storage shelf	**G**	Control panel
D	AC power cord	**H**	Printer paper drawer

CAUTION Always lock the wheel brake (**F**) when the cart is not in use. Press down on the wheel brake to set or to release the wheel brake.

Figure 1-4 PageWriter Trim I Cardiograph (Rear View)



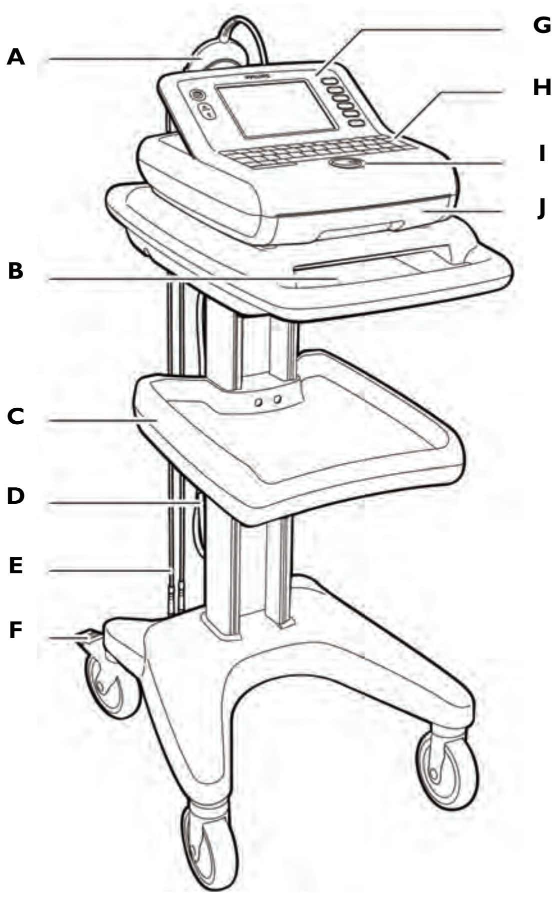
I	AC power cord connector	**L**	Battery door
J	Equipotential grounding post	**M**	PIM connector
K	Fuse door	**N**	Serial connector (not supported)

WARNING **Do not connect a LAN cable connector to the PIM connector. Do not plug a telephone connector into the PIM connector.**

PageWriter Trim II and III Cardiograph Parts

The following sections show front, side, and rear views of the PageWriter Trim II and III cardiographs.

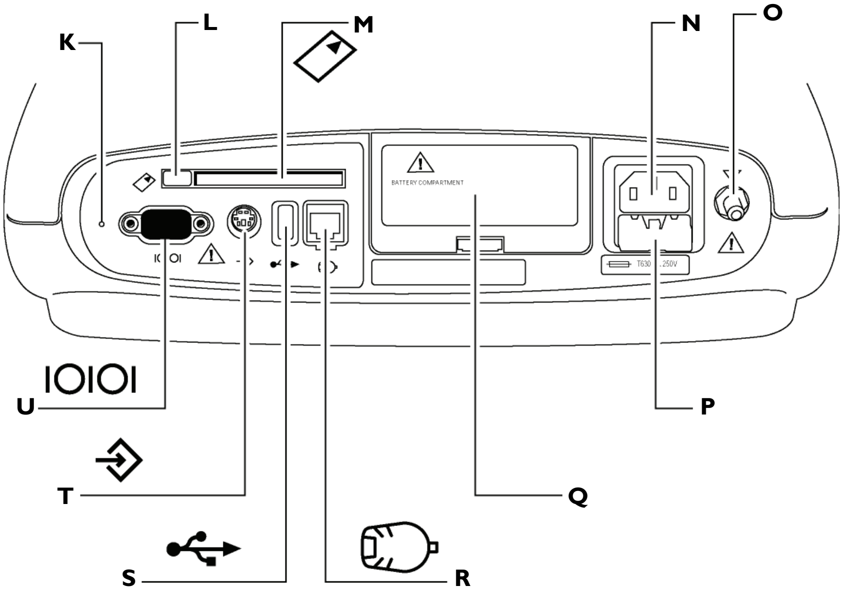
Figure 1-5 PageWriter Trim II and III Cardiograph and Cart (Front View)



A	Patient Interface Module (PIM)	**F**	Wheel brake
B	Printer paper/report storage slot	**G**	Control panel
C	Storage shelf	**H**	Keyboard
D	AC power cord	**I**	Trim knob
E	PIM leads	**J**	Printer paper drawer

CAUTION Always lock the wheel brake (**F**) when the cart is not in use. Press down on the wheel brake to set or to release the wheel brake.

Figure 1-6 PageWriter Trim II and III Cardiograph (Rear View)



K	Reset button	**Q**	Battery door
L	PC card eject button	**R**	PIM connector
M	PC card slot	**S**	SmartCard reader or USB memory stick connector
N	AC power connector	**T**	Barcode reader or magnetic card reader connector
O	Equipotential grounding post	**U**	Serial connector (not supported)
P	Fuse door		

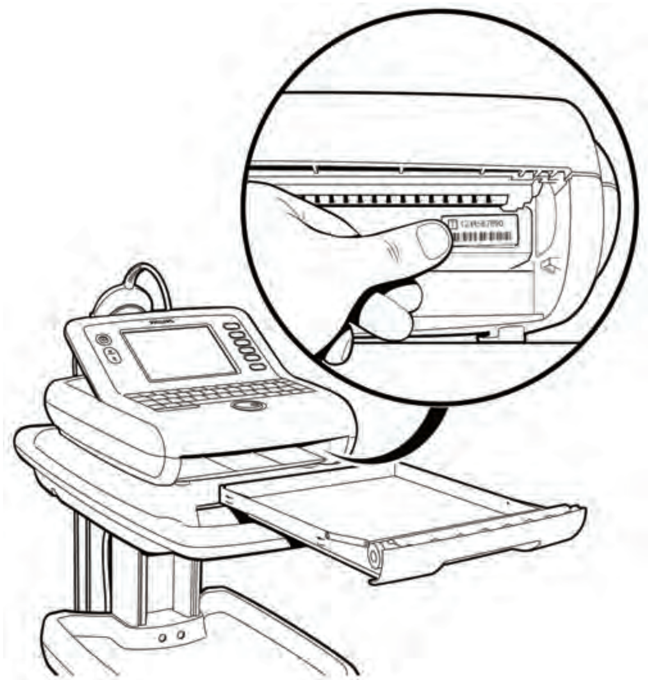
WARNING **Do not connect the LAN cable connector into the PIM connector. Do not plug a telephone connector into the PIM connector.**

PageWriter Trim II and III Token Label

A token label is affixed to the interior of the cardiograph, above the paper tray. To locate the token label, remove the paper tray from the cardiograph. The label is located on the far right side of the metal housing. The number on the token label is referred to when the cardiograph is serviced, software is installed, or when an optional feature is added to the cardiograph. Ensure that the most current token label is affixed to the cardiograph. For more information on the token label, contact the nearest Philips Response Center, see page 1-50 for more information.

Figure 1-7 PageWriter Trim Cardiograph Token Label

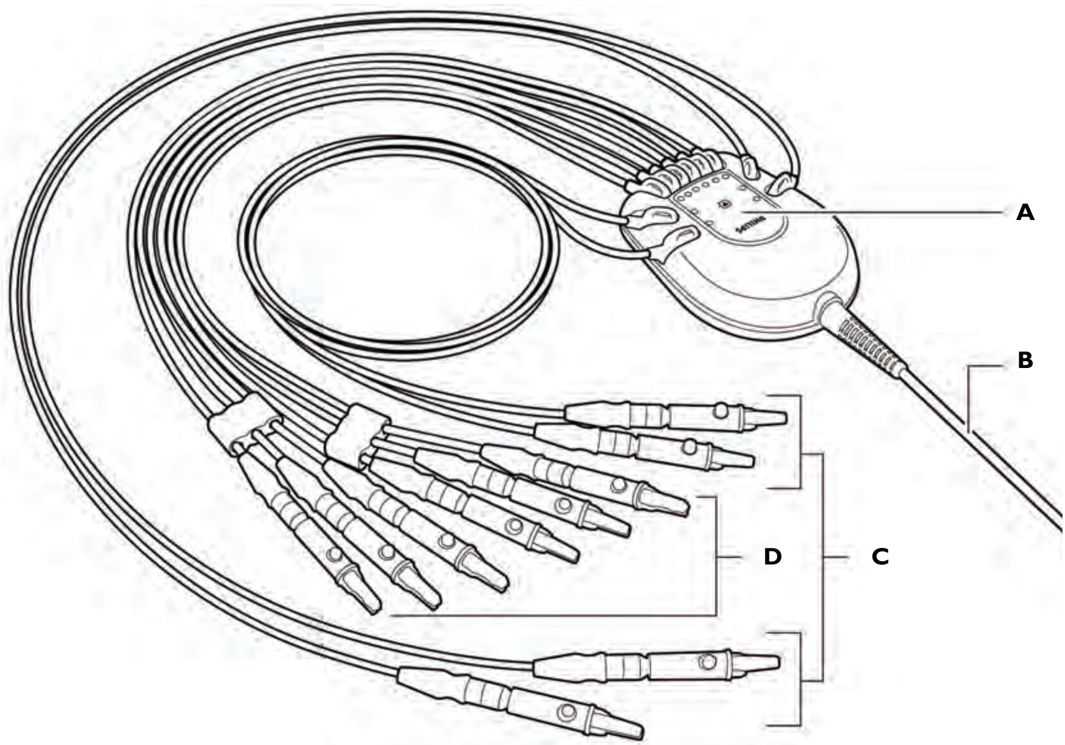
Figure 1-8 Location of Token Label on Cardiograph



Patient Interface Module (PIM)

The Patient Interface Module (PIM) is a hand-held device that connects to the cardiograph. The lead wires on the PIM attach to the electrodes placed on the patient. The exterior of the PIM is labeled and color-coded for quick and easy lead identification.

Figure 1-9 Patient Interface Module



A Lead wire labeling

B Patient data cable

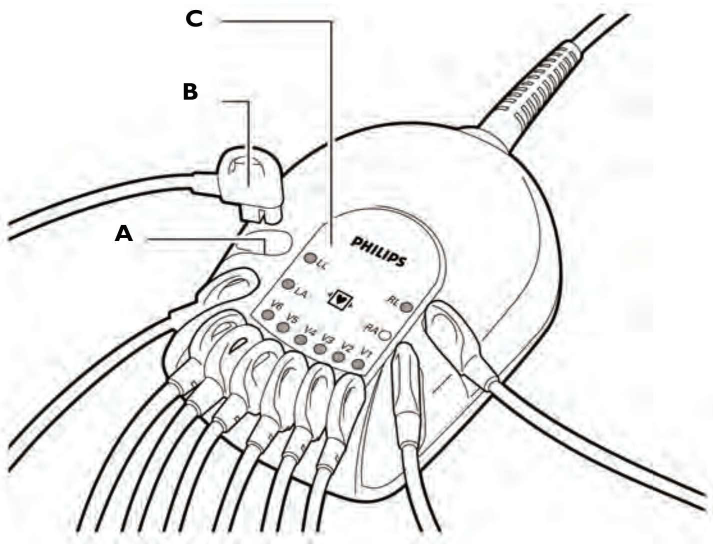
C Limb lead wires

D Precordial (chest) lead wires

Inserting the Lead Wires into the PIM

The exterior of the PIM is color-coded for quick and easy lead identification. The lead wires that are included with the cardiograph must be inserted into the correct lead wire connector on the PIM.

Figure 1-10 PIM connector and lead wire labeling



A Lead wire connector

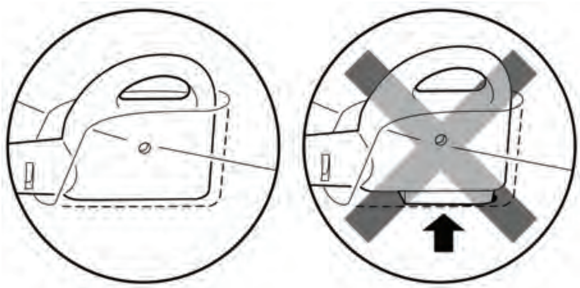
B Lead wire

C Lead wire labeling

To insert the lead wires into the PIM:

1. Locate the lead wires in the cardiograph packaging.
2. Match the lead wire to the correct lead wire connector on the PIM.
3. Insert each lead wire into the correct lead wire connector on the PIM.

4 Push down on the lead wire to ensure that there is no gap between the lead wire and the connector.

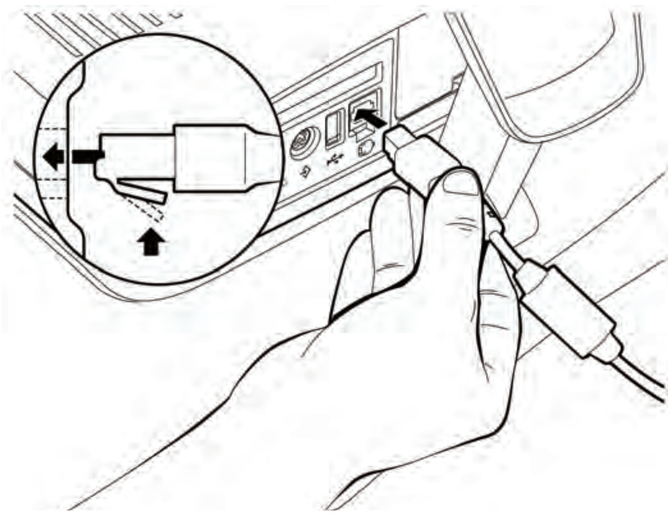


Connecting the PIM to the Cardiograph

Connect the patient data cable on the PIM to the PIM connector port () on the rear panel of the cardiograph.

To connect the PIM to the cardiograph:

▶ Connect the patient data cable to the PIM connector port on the rear of the cardiograph.

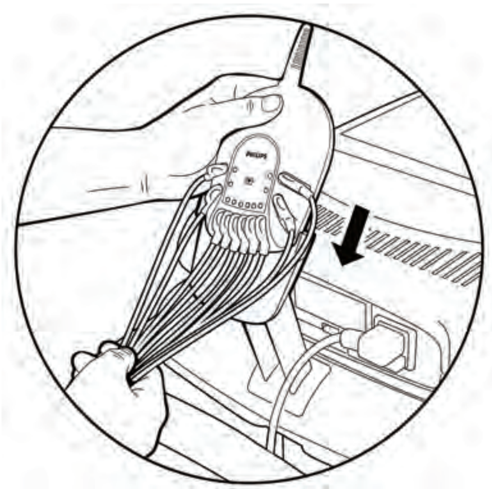


WARNING **To ensure safety and prevent damage to the system, only connect the patient data cable to the PIM connector port on the rear of the cardiograph.**

Placing the PIM in the Holder

Place the PIM in the holder when not in use. The PIM holder is located on the rear of the optional cardiograph cart. Ensure that the lead wires are facing down.

Figure 1-11 Inserting the PIM into the holder

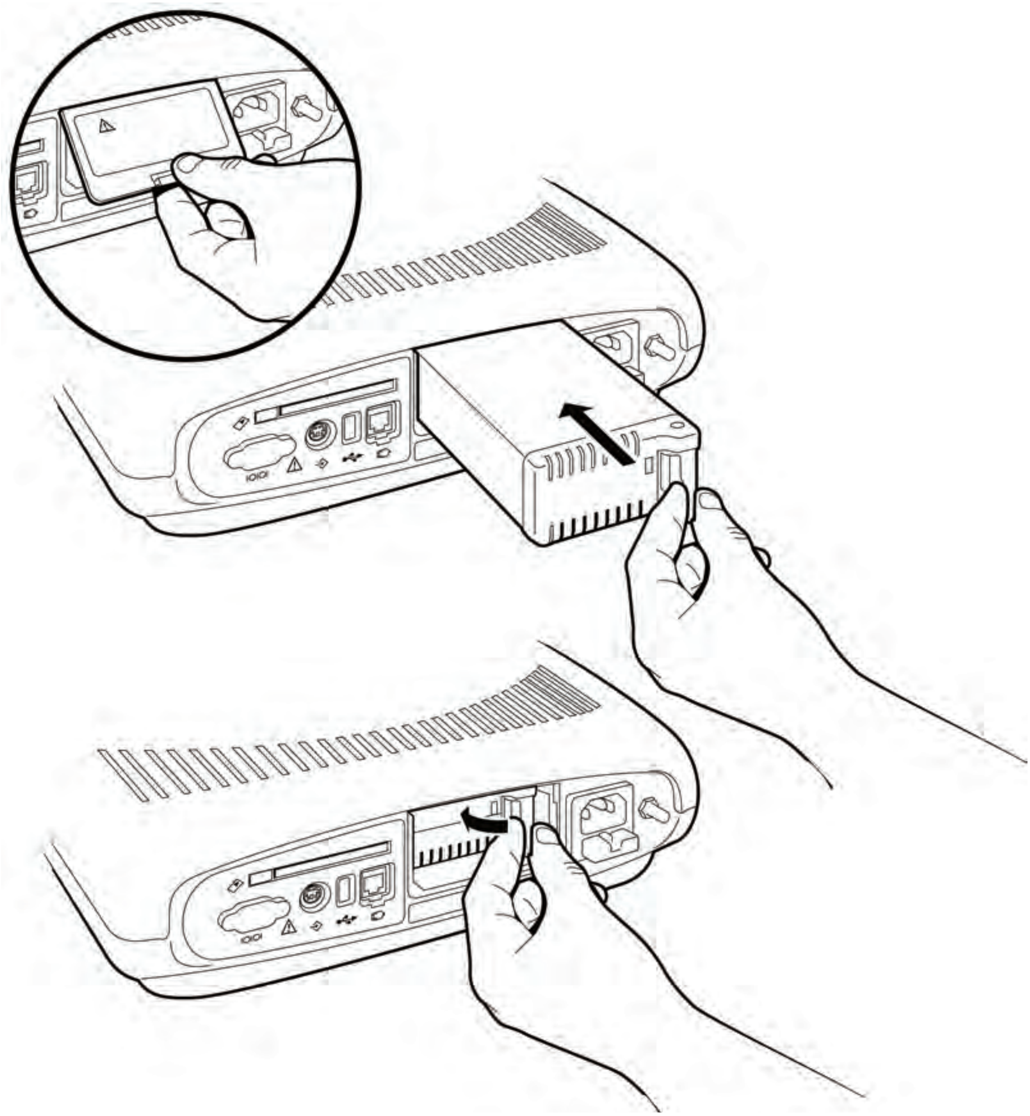


CAUTION Ensure that the PIM patient data cable and lead wires do not drag on the ground or become tangled in the cart wheels.

Installing the Battery

The cardiograph is shipped with one removable battery. Install the battery before plugging the cardiograph into AC power.

Figure 1-12 Installing the Battery



To install the battery:

1. Open the battery door on the rear of the cardiograph. Push the recessed tab in and pull it down to open the battery door.
2. Insert the battery into the compartment with the connector facing down and the pull tab facing out.
3. Push in the pull tab to lock it.
4. Reattach and close the battery door.
5. Attach the AC power cord to the rear of the cardiograph.
6. Plug the AC power cord into a grounded electrical outlet. Check that the green AC power indicator light is on (front of cardiograph).
7. Charge the battery for at least twenty-four hours before mobile use.

Charging the Battery

After the battery is installed in the cardiograph, it must be charged for at least twenty-four hours prior to use.

With the battery fully charged, the cardiograph will print up to thirty Auto ECGs or provide thirty minutes of continuous Rhythm printing.

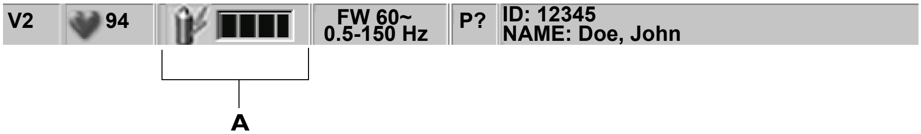
The green AC indicator light on the front of the cardiograph indicates that the battery is charging when the cardiograph is plugged into AC power. Plug the cardiograph into AC power whenever possible.

To disconnect the cardiograph from AC power, unplug the AC power cord from the grounded electrical outlet.

PageWriter Trim II and III Battery Charging Indicator

On the PageWriter Trim II and III, a lightening bolt icon () appears on the Battery Level Indicator on the Status Bar to indicate that the cardiograph is plugged into AC power and the battery is charging. Check the Battery Power Indicator on the Status Bar to ensure that the battery is fully charged. The cardiograph can operate on AC power while the battery is charging, but the battery will charge at a slower rate.

Figure 1-13 PageWriter Trim II and III Battery Level Indicator



A Battery level indicator

Table 1-1 PageWriter Trim II and III Battery Level Indicator

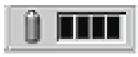
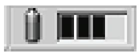
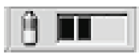
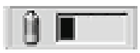
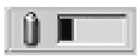
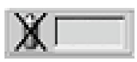
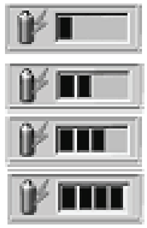
Battery Level	Status Indicator
Fully Charged Battery	
75% power capacity	
50% power capacity	
Low Battery Power	
Low Battery Alert; alternate icons flashing, cardiograph *beeps* until the unit is plugged into AC power	 
No or Dead Battery	

Table 1-1 PageWriter Trim II and III Battery Level Indicator *(continued)*

Battery Level	Status Indicator
Battery is charging (moving bars)	

PageWriter Trim I Battery Charging Indicator

On the PageWriter Trim I, the battery icon () flashes on the LCD display when the cardiograph is plugged into AC power and the battery is charging, see “LCD Display” on page 1-40.The battery level indicator displays the following battery icon () when the battery is fully charged. The cardiograph can operate on AC power while the battery is charging, but the battery will charge at a slower rate.

Table 1-2 PageWriter Trim I Battery Level Indicator

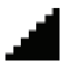
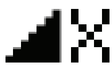
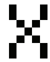
Battery Level	Status Indicator
Fully Charged Battery	
50% power capacity	
Low Battery Power	
Low Battery Alert	

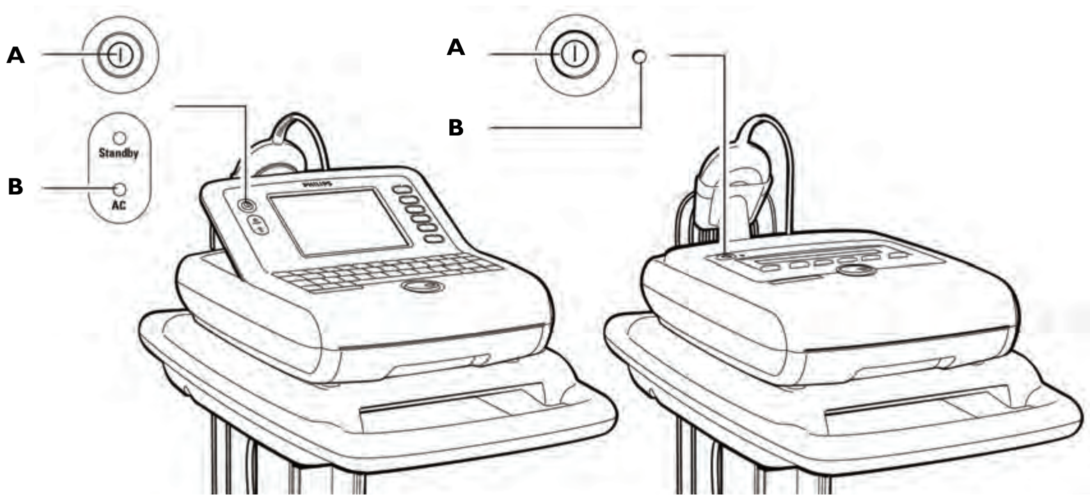
Table 1-2 PageWriter Trim I Battery Level Indicator *(continued)*

Battery Level	Status Indicator
No or Dead Battery	
Battery is charging	

Powering on the Cardiograph

Ensure that the battery is inserted into the cardiograph before turning on the AC power to the cardiograph. See “Installing the Battery” on page 1-17. The cardiograph is powered by the battery or by AC power. The battery must be inserted into the cardiograph at all times. The cardiograph printer will not function if the battery is not installed.

Figure 1-14 The On/Standby Button



A On/Standby button (PageWriter Trim II/III) *or* On/Off button (PageWriter Trim I)

B AC Power Indicator Light

To turn on the cardiograph:

1. Plug the AC power cord into the AC power connector on the rear of the cardiograph.
2. Plug the AC power cord into a grounded electrical outlet with appropriate electrical ratings. The AC power indicator light on the front panel of the cardiograph illuminates when the cardiograph is plugged into AC power.
3. Press the On/Standby or On/Off button on the front of the cardiograph.

Using the On/Standby Button

PageWriter Trim II and III only

The On/Standby button is used to put the cardiograph into Standby and to shut down the cardiograph.

Table 1-3 Standby and Shut Down Setting Description

Setting	Description
Standby	■ Use between brief periods of inactivity to save battery power ■ The cardiograph is in a *sleep* mode ■ The display screen is black ■ Patient ID information is not saved when the cardiograph is put into Standby ■ Temporary report settings are saved when the cardiograph is put into Standby ■ The battery charges when the cardiograph is plugged into AC power ■ The green Standby indicator light (front of cardiograph) is lit ■ Press any key on the keyboard to return the cardiograph to active use ■ The cardiograph returns to the previously viewed screen when taken out of Standby

Table 1-3 Standby and Shut Down Setting Description *(continued)*

Setting	Description
Shut Down	■ Use for extended periods of cardiograph inactivity ■ The cardiograph is shut down ■ The display screen is black ■ The battery charges when the cardiograph is plugged into AC power ■ Press the On/Standby button to restart the cardiograph

TIP There are configurable features on the cardiograph that can be used to help prolong battery life (see page 2-33). Proper battery care and maintenance (including frequent and full charging of the battery) will also help to prolong battery life.

TIP Plug the cardiograph into AC power whenever possible to help prolong battery life.

To put the cardiograph into Standby:

▶ Press the On/Standby button. A message appears that the cardiograph is entering Standby. The green Standby indicator light (front of cardiograph) illuminates. The battery charges when the cardiograph is plugged into AC power.

NOTE The cardiograph cannot be put into Standby if the Patient ID screen is open or if the cardiograph is printing an ECG.

To exit Standby:

▶ Press the On/Standby button. The splash screen appears. The R/T (real-time) ECG screen appears.

To shut down the cardiograph:

▶ Press and hold the On/Standby button for three seconds then release it. A message appears that the cardiograph is shutting down.

To turn on the cardiograph:

▶ Press the On/Standby button. The cardiograph displays a series of startup screens and then the R/T (real-time) ECG screen displays.

Loading Printer Paper

Replace the printer paper when a red stripe appears on the printed ECG report. Only use Philips Medical Systems replacement printer paper. For part number and ordering information, see page 1-46.

Tips for loading printer paper

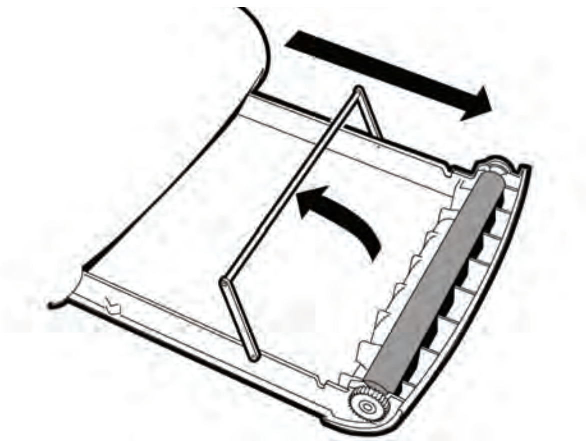
- Always load less than 100 sheets of printer paper into the paper tray
- Ensure that the entire first page of the new paper roll is fully draped over the roller before closing the printer door

PageWriter Trim II and III:

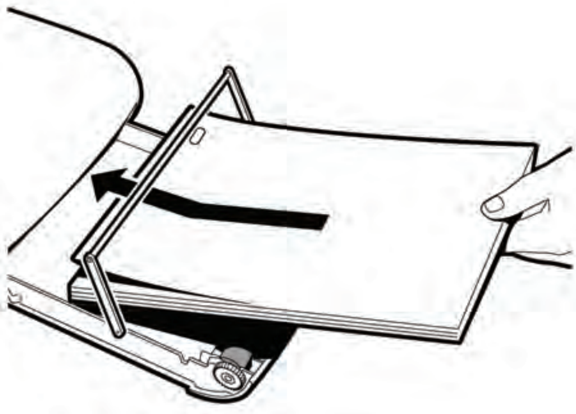
- Ensure that the paper size configured for the cardiograph is the same size paper being loaded into the paper drawer (see page 2-33)

To load paper into the printer:

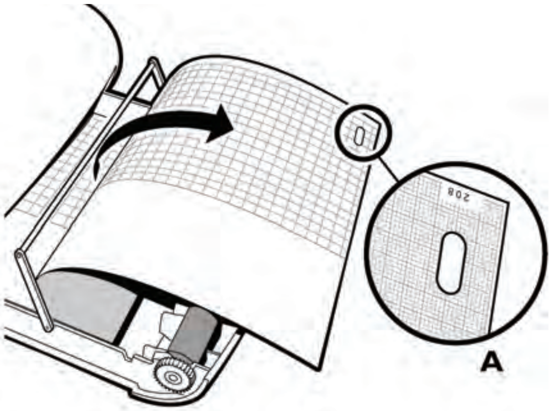
1 Open the paper drawer on the front of the cardiograph and lift up the bar.



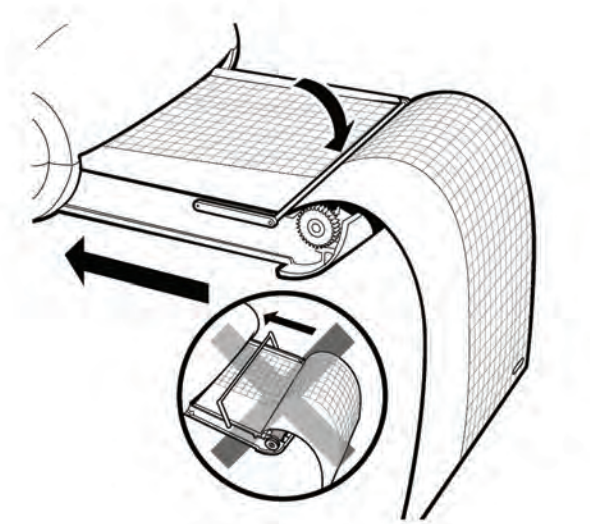
2 Insert a new pack of printer paper as shown. Ensure that no more than 100 sheets are being inserted into the paper tray.



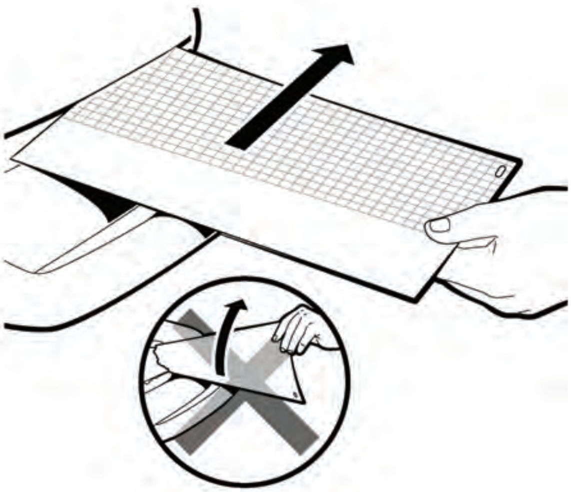
3 Ensure that the paper sensor hole (**A**) is positioned as shown.



4 Drape the entire first sheet over the roller. Ensure that the perforated edge of the paper aligns with the edge of the paper drawer. Lower the bar onto the paper and close the drawer.



5 Tear off the first sheet as shown.



Using the PC Card Slot

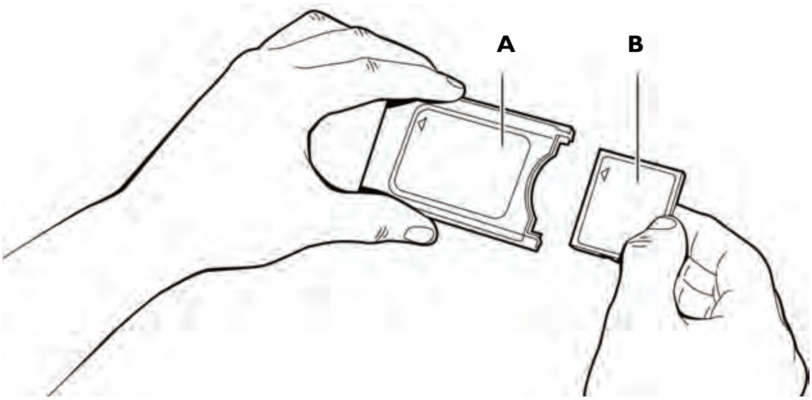
PageWriter Trim II and III only

The PC card slot (rear of cardiograph) is used with the optional PC card, wireless LAN card, LAN card, or modem card. All of these are optional accessories available with the cardiograph, and are inserted into the PC card slot. For information on ordering any of these accessories, see “Supplies and Ordering Information” on page 1-46.

Using the PC Card

The PC card contains a 128 MB removable memory card that can be used to transfer ECGs to a TraceMasterVue ECG Management System, or to transfer Orders to the cardiograph. The PC card can store up to 150 ECGs or Orders. The PC card can also be used to store a custom settings file. This custom settings file can be used to transfer custom settings to additional cardiographs, or can be used to restore custom settings on the cardiograph. For information on transferring ECGs or Orders to the cardiograph, see Chapter 4 “PageWriter Trim II and III Orders and Archive” of the *PageWriter Trim Instructions for Use* on the User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com). For information on using the PC card to save custom configuration settings, see page 2-64.

Figure 1-15 The PC card

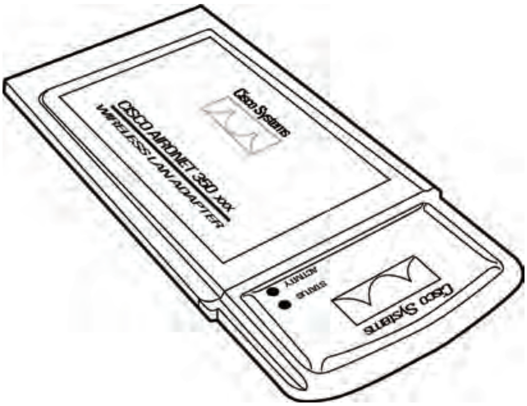


A PC card adapter **B** 128 MB removable memory card

Using the Wireless LAN Card

The PageWriter Trim cardiograph offers a wireless LAN option for the purpose of transferring ECGs to a TraceMasterVue ECG Management System. The wireless LAN connection is compatible with the 802.11b wireless standard. To install and configure the wireless LAN card, see the *PageWriter Trim Cardiograph Wireless LAN Installation Instructions* available on the User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com).

Figure 1-16 Wireless LAN Card

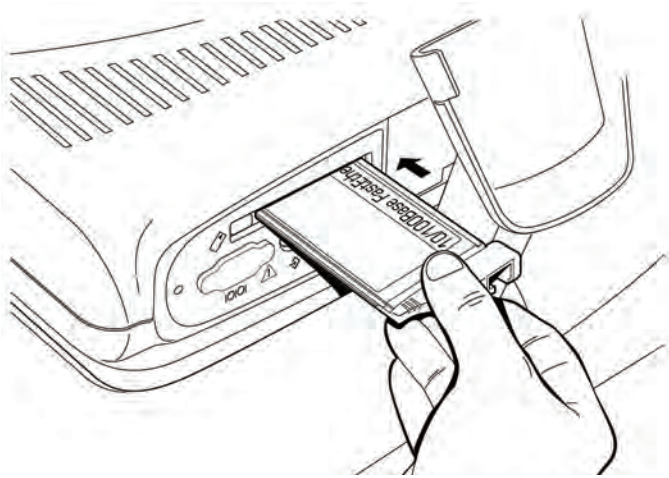


Using the LAN Card

The LAN card is used to transfer ECGs to a TraceMasterVue ECG Management System through a 10/100-Base T Ethernet network connection. The LAN card must be configured before using it with the cardiograph, see page 2-53. For information on transferring ECGs over a network connection to a TraceMasterVue ECG Management System, see Chapter 4 “PageWriter Trim II and III Orders and Archive” of the *PageWriter Trim Instructions for Use* on the User Documentation CD or download the file from the Philips InCenter web site (incenter.medical.philips.com).

CAUTION Only use the shielded LAN cable provided with the PageWriter Trim cardiograph. Do not use LAN cables with the cardiograph that have not been approved by Philips Medical Systems.

Figure 1-17 Inserting LAN card into PC Card slot

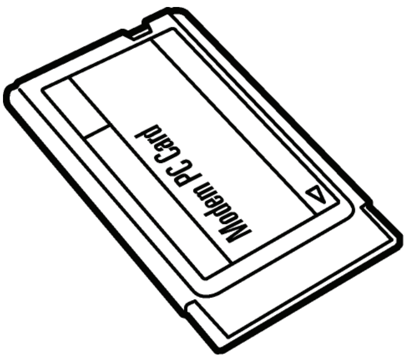


Using the Modem Card

The modem card is used to fax ECGs to a remote receiving station, or to transfer ECGs by modem to a TraceMasterVue ECG Management System. The modem card used either as a fax or as a modem must be configured before initial use with the cardiograph, see page 2-45.

For information on faxing ECGs or transferring ECGs by modem, see Chapter 4 “PageWriter Trim II and III Orders and Archive” of the *PageWriter Trim Instructions for Use* on the User Documentation CD or download the file from the Philips InCenter web site (incenter.medical.philips.com).

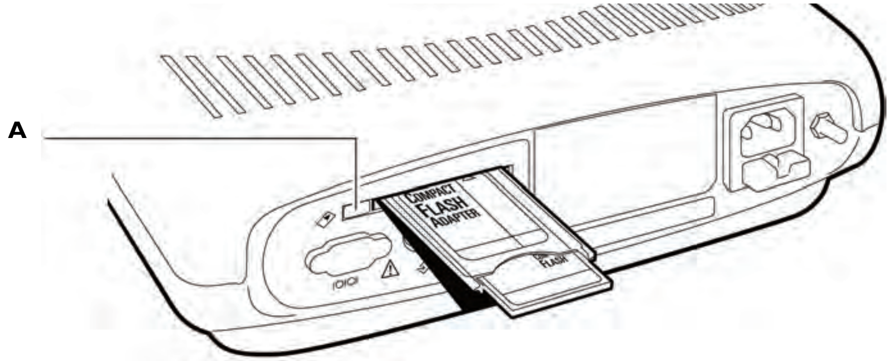
Figure 1-18 The Modem Card



Installing and Removing a PC Card

The PC card, wireless LAN card, LAN card, or modem card are all inserted into the PC card slot.

Figure 1-19 Removing the PC card from the cardiograph



A PC card eject button

To insert or remove a card from the PC card slot:

1. Insert the card into the PC card slot (rear of cardiograph) and gently push it in.
2. Push the PC card eject button to the left of the card slot to eject the card.
3. Pull out the card.

Using the USB Memory Stick

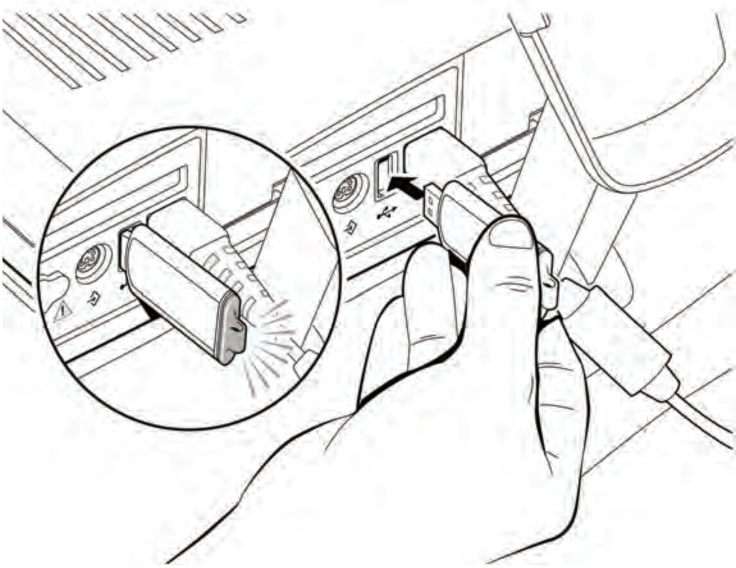
PageWriter Trim II and III only

The USB memory stick is an optional accessory that is used to transfer ECGs to a TraceMasterVue ECG Management System, or to transfer Orders to the cardiograph. The USB memory stick can store up to 150 ECGs or Orders.

The PageWriter Trim cardiograph only supports the USB memory stick that is shipped with the cardiograph, or that is available for purchase as an optional accessory from Philips Medical Systems. Philips does not guarantee that other USB memory sticks are compatible with the PageWriter Trim cardiograph.

The memory stick is inserted into the USB connector on the rear of the cardiograph. The memory stick illuminates when it is properly inserted into the USB connector.

Figure 1-20 The USB memory stick



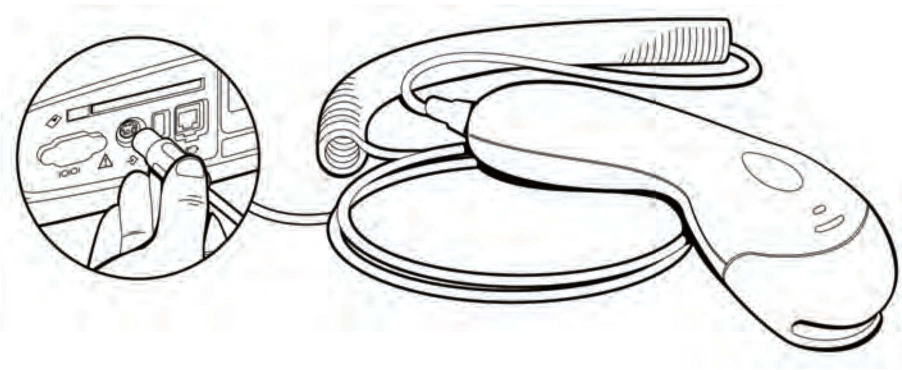
Using the Barcode Reader

PageWriter Trim II and III only

The barcode reader is an optional accessory available with the cardiograph. The barcode reader is used to quickly enter Patient ID information by scanning a barcode.

The barcode reader attaches to the barcode reader connector ( ) on the rear panel of the cardiograph.

Figure 1-21 The Barcode Reader



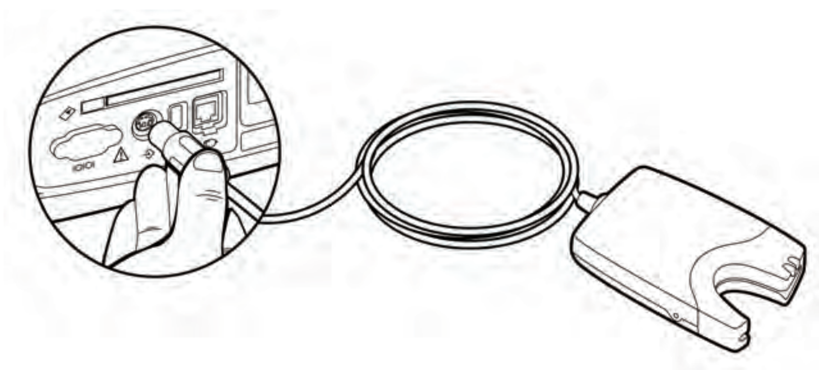
Using the Magnetic Card Reader

PageWriter Trim II and III only

The magnetic card reader is an optional accessory available with the cardiograph. The magnetic card reader is used to quickly enter Patient ID information. The Patient ID fields that are entered when a magnetic card is scanned are configurable, see page 2-40.

The magnetic card reader attaches to the magnetic card reader connector ( ) on the rear panel of the cardiograph. For information on ordering the magnetic card reader, see "Supplies and Ordering Information" on page 1-46.

Figure 1-22 The Magnetic Card Reader



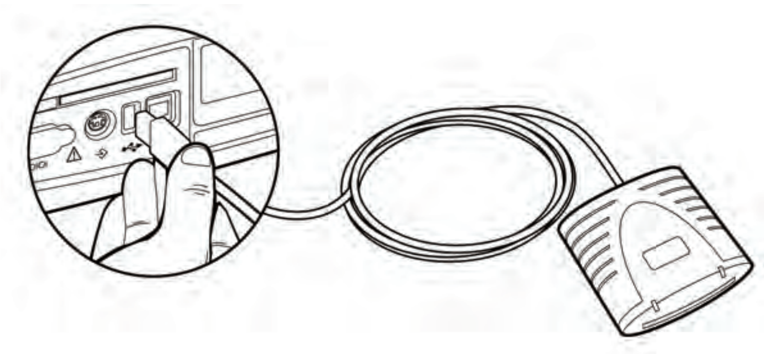
Using the SmartCard Reader

PageWriter Trim II and III only

The SmartCard Reader is an optional accessory available with the cardiograph. The SmartCard Reader is used to quickly enter Patient ID information.

The SmartCard Reader attaches to the USB connector (•⟷) on the rear panel of the cardiograph. For information on ordering the SmartCard Reader, see “Supplies and Ordering Information” on page 1-46.

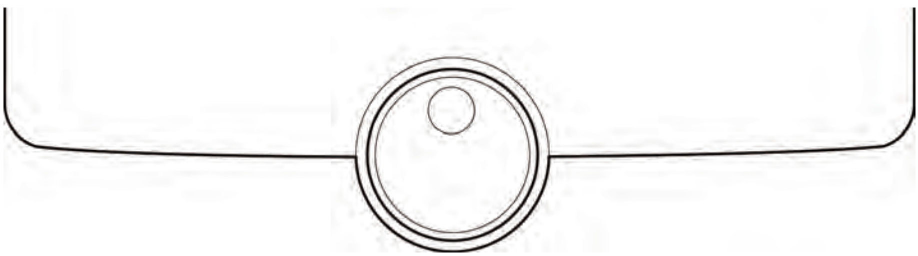
Figure 1-23 The SmartCard Reader



Using the Trim Knob

All models of the PageWriter Trim cardiograph have a Trim Knob that is used to scroll through and select items on any display menu.

Figure 1-24 The Trim Knob



To use the Trim Knob:

1. Turn the Trim Knob left or right to scroll through the selections in a menu list.
2. Push the Trim Knob to select an item on the menu.

Navigation Tips

PageWriter Trim II and III only

In addition to the Trim Knob, individual keys on the keyboard may be used to navigate on the software screens.

Opening and Closing Screens

To open and close any screen:

1. Press the *Enter* key () to close any screen or window and save the information.
2. Press the *Esc* key (

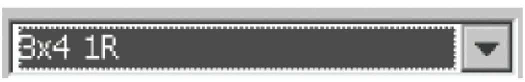
) to exit any screen without saving information.

Opening and Closing Menus

To open and close menus:

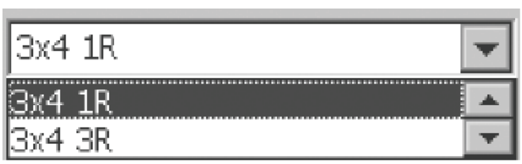
1. Press the *Tab* key () to highlight a drop-down menu (in blue).

Figure 1-25 Highlighted drop-down menu



2. Press the space bar () to open the drop-down menu.

Figure 1-26 Opened drop-down menu

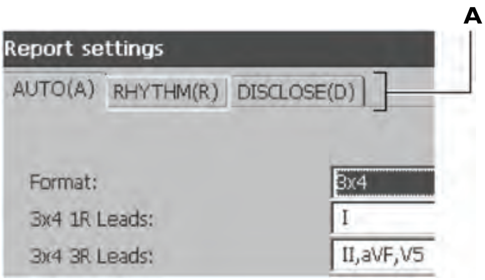


3 Press the up or down arrow key () to scroll through the selections in the menu.

4 Press the space bar to select an item and to close the drop-down menu.

Selecting Tabs

Figure 1-27 Tabs on Patient ID screen



A Tabs

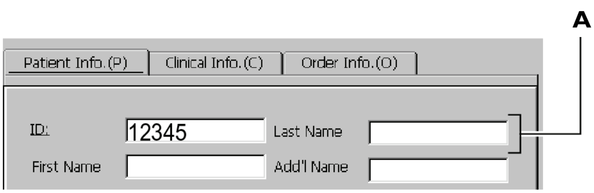
To select a tab:

1 Press and hold the *Alt* key (
) on keyboard.

2 Press the letter key in parenthesis on the tab. For example, to select the **Rhythm** tab, press the *Alt* key (
) and the *R* (
) key.

Entering Information on a Screen

Figure 1-28 Information fields on the Patient ID screen



A Field

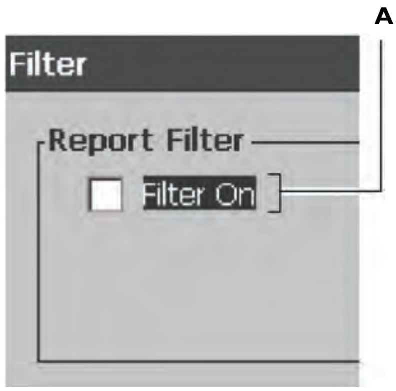
To enter information on a screen:

1 Press the *Tab* key ( ) to go to the next field (white square) on any screen.

2 Press the *Tab* key and the *Shift* key ( ) at the same time to go to the previous field (white square) on any screen.

Selecting Check Boxes

Figure 1-29 Check box on the Filter screen



A Highlighted check box

To select a check box:

1 Press the *Tab* key ( ) to highlight the text next to the check box.

2 Press the space bar ( ) to select the check box. A check mark ( ) appears in the box.

3 Press the *Enter* key ( ) to save the information and to exit the screen.

4 Press the *Esc* key ( ) to close the screen without saving the information.

PageWriter Trim I Features

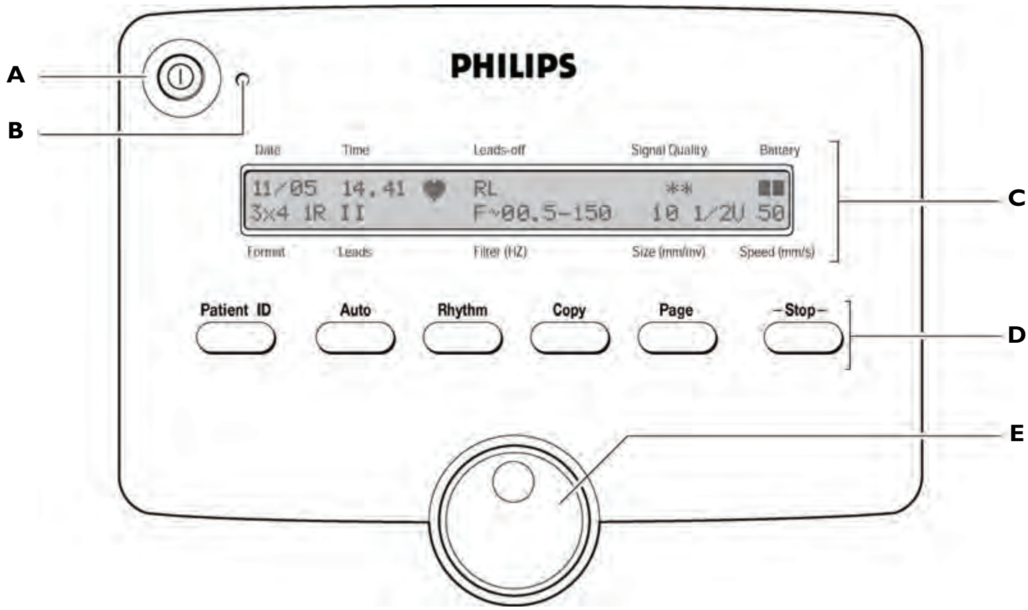
The PageWriter Trim I has an LCD display that shows all operating information. The buttons on the main panel of the cardiograph are used for all cardiograph functions. Simply select the buttons on the main panel to use the different functions on the cardiograph.

NOTE Complete the Interactive Training Program on CD before operating the cardiograph. The Interactive Training program includes instructional information on what an ECG is, patient preparation techniques, how to operate the cardiograph, and quality assessment techniques to ensure high-quality ECGs. For information on using the training program, see page 1-4.

Control Panel

The control panel contains all of the main functions on the cardiograph. For more information on any feature, see Chapter 3 “The Patient Session” of the *PageWriter Trim Instructions for Use* on the User Documentation CD or download the file from the Philips InCenter web site (incenter.medical.philips.com).

Figure 1-30 PageWriter Trim I Control Panel



A On/off button

B AC power on indicator light

C LCD Display

D Function buttons

E Trim Knob

LCD Display

The LCD display shows the current settings on the cardiograph.

Figure 1-31 PageWriter Trim I LCD Display

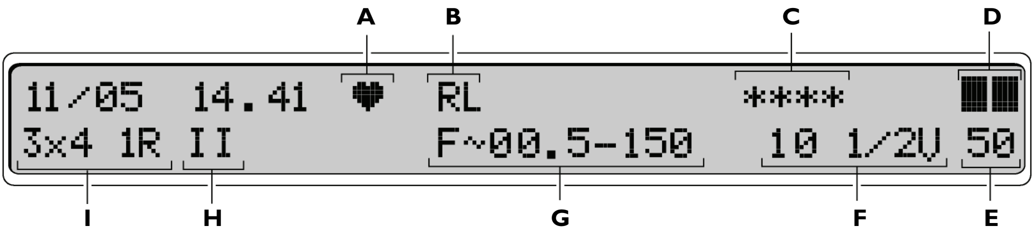


Table 1-4 PageWriter Trim I LCD Display

Label	Description
A	Heart Beat Detector
B	Leads Off Indicator
C	Signal Quality Indicator
D	Battery Level Indicator
E	Waveform Speed (mm/sec)
F	Waveform Size (mm/mV)
G	Filter Settings
H	Rhythm Lead
I	Report Format

Function Buttons

All of the functions on the PageWriter Trim I cardiograph are controlled by the function buttons located on the main panel of the cardiograph.

Figure 1-32 Function Buttons on PageWriter Trim I cardiograph

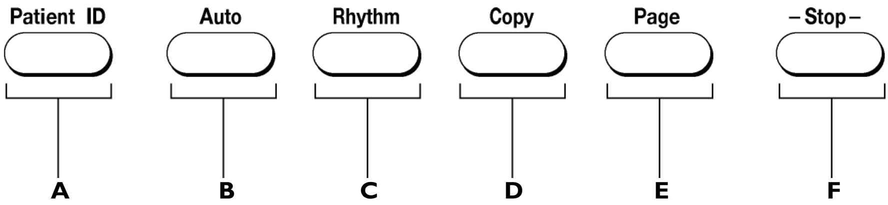


Table 1-5 Function Button description

Label	Button	Push button to...
A	Patient ID	Enter Patient ID information (age, gender)
B	Auto	Take a 3x4, 1R or 6x2 Auto ECG
C	Rhythm	Print out a continuous Rhythm report (3 or 6 leads) until the Stop button (**F**) is pressed
D	Copy	Print out a copy of the last recorded Auto 12-lead ECG
E	Page	Advance the printer paper to the beginning of the next page
F	Stop	■ Stop the printing of a Rhythm report ■ Return to the previous screen on the LCD display

PageWriter Trim II and III Features

For more information on any feature, see Chapter 3 “The Patient Session” of the *PageWriter Trim Instructions for Use* on the User Documentation CD or download the file from the Philips InCenter web site at (incenter.medical.philips.com).

Figure 1-33 PageWriter Trim II and III Control Panel

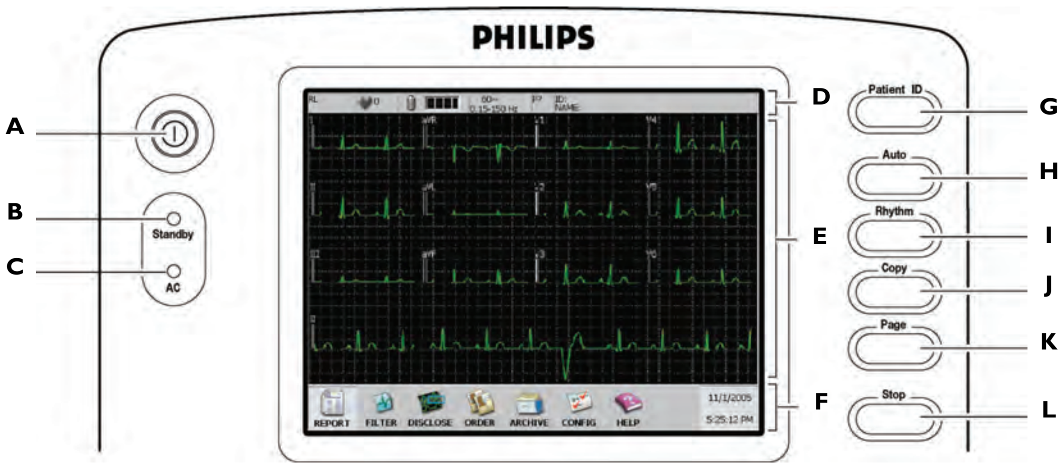


Table 1-6 PageWriter Trim II and III Control Panel

Label	Feature Name	Description
A	**On/Standby button**	■ Push to turn on the cardiograph or to put the cardiograph into Standby (power save) when the cardiograph is not in use. ■ Push and hold for three seconds, then release to shut down the cardiograph for extended periods of inactivity.
B	**Standby indicator**	When lit, the cardiograph is in Standby (power save) mode and the screen is black.
C	**AC Power Indicator**	When lit, the cardiograph is plugged into AC power and the battery is charging.

Table 1-6 PageWriter Trim II and III Control Panel *(continued)*

Label	Feature Name	Description
D	**Status Bar**	Displays current cardiograph settings including: leads off, battery level indicator, patient heart rate, filter settings, pacing detection settings, and patient name and ID.
E	**Waveform Display**	■ Displays real-time waveforms (waveforms are color-coded to indicate signal quality on the PageWriter Trim III). ■ Press the up  or down  arrow key to change the lead format displayed on the screen ■ Available lead formats include: 6x2, 3x4 1R, 3x4, and 3 lead formats
F	**Command Toolbar**	Buttons contain the major functions of the cardiograph, with the current date and time at the right.
G	**Patient ID button**	Press button to enter Patient ID information at the beginning of a patient session.
H	**Auto button**	■ Press button to print out an Auto 12-lead ECG. ■ Press button twice to print a STAT 12-lead ECG.
I	**Rhythm button**	Press button to print out a continuous Rhythm report (up to 12 leads) until the Stop button (**L**) is touched.
J	**Copy button**	Press button to print out a copy of the last recorded Auto 12-lead ECG.
K	**Page button**	Press button to advance the printer paper to the beginning of the next page.
L	**Stop button**	Press button to stop the printing of a Rhythm report or any other printing function.

PageWriter Trim II and III Status Bar

The Status Bar includes all current operating information for the cardiograph.

Figure 1-34 Status Bar

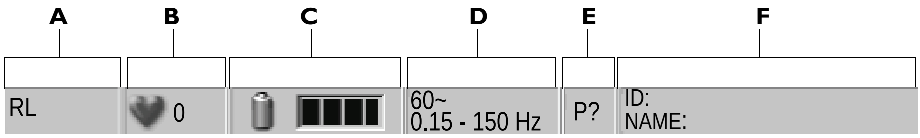


Table 1-7 Status Bar Description

Label	Feature Name	Description
A	**Leads Off Indicator**	Displays the name of any loose or disconnected leads or electrodes.
B	**Heart Rate Indicator**	Displays the current patient heart rate in beats per minute.
C	**Battery Level Indicator**	Displays the current battery level and if the batteries are charging.
D	**Filter Settings**	Displays the filter settings for the current patient session.
E	**Pacing Detection Settings**	Displays the Pacing Detection Setting used for the current patient session.
F	**Patient Name and ID**	Displays the patient name and ID for the current patient session.

PageWriter Trim II and III Command Toolbar

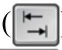
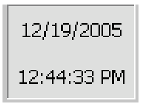
The Command Toolbar includes all of the main functions of the cardiograph. Press the *Tab* key () or turn the Trim Knob to highlight a button. Press the space bar () or press the Trim Knob to select the button.

Figure 1-35 Command Toolbar

Table 1-8 Command Toolbar

Button	Select this Button to...
	■ Change Auto ECG settings ■ Change pacing detection settings ■ Change Rhythm report settings ■ Change the lead on a Disclose report
	Turn on or turn off optional filters (Baseline Wander, Artifact)
	Print a report of one minute of continuous waveform data for one selected lead
	■ Load Orders on the cardiograph from a PC card or a USB memory stick ■ Enter an Order using the keyboard, or edit or delete Orders
	■ Review, edit, or print saved 12-lead ECGs ■ Transfer 12-lead ECGs to a TraceMasterVue ECG Management System ■ Download ECGs from a TraceMasterVue Remote Site and view and print ECGs on the cardiograph

Table 1-8 Command Toolbar *(continued)*

Button	Select this Button to...
	■ Configure the cardiograph or change configured settings ■ Change the access passwords for features on the cardiograph ■ Save all cardiograph configured settings to a PC card or to a USB memory stick in case of system failure ■ Save all cardiograph configured settings to a PC card or to a USB memory stick in order to upload the settings to multiple cardiographs
	Get Help when using the cardiograph
	Change the current date and time

Supplies and Ordering Information

The part numbers for all supplies for the PageWriter Trim cardiograph are listed in this section.

Ordering Supplies

All supplies may be ordered on the web at:

http://shop.medical.philips.com

Use the part numbers listed below for reference to ensure that the correct supplies are ordered.

PageWriter Trim Cardiograph Supply Part Numbers

Patient Interface Module (PIM)

Part Number	Description
989803130091	Patient Interface Module (AAMI)
989803130101	Patient Interface Module (IEC)

Replacement Fuse

Part Number	Description
453564000371	1.5 amp (250 V) Time-Delay Fuse

Complete Lead Sets

Part Number	Description
989803129161	Complete Lead Set (AAMI)
989803129191	Complete Lead Set (IEC)

Replacement Lead Sets and Accessories

Part Number	Description
989803129141	Limb Lead Set, 99 cm/39 in (AAMI)
989803129151	Chest Lead Set, 61 cm/24 in (AAMI)
989803129171	Limb Lead Set, 99 cm/39 in (IEC)
989803129181	Chest Lead Set, 61 cm/24 in (IEC)
989803129201	Long Limb Lead Set, 137 cm/54 in (IEC)

Replacement Lead Sets and Accessories *(continued)*

Part Number	Description
989803129211	Long Chest Lead Set, 99 cm/39 in (IEC)
989803129221	Long Complete Lead Set (IEC)

Electrodes

Part Number	Description
13943B	Disposable cardiography electrode, resting diagnostic ECG
M2253A	Disposable electrode, adult, resting ECG (not available in Japan)
40420A	Disposable electrode, diagnostic, pre-gelled
13944B	Wet Gel Foam Electrode, resting ECG
989803129231	Alligator Clips for Disposable Tab Electrodes (AAMI)
989803129241	Alligator Clips for Disposable Tab Electrodes (IEC)
M2254A	Tab electrode adapter
40431B	Alligator Clip Adapter for 1/8 post
40498E	Snap Lead Electrode (IEC)
40475A	Snap Lead Electrode Adapter
40490E	15mm Suction Electrode
40421A	Push-in Connect Welsh Electrode
40494E	Limb Clamp Electrode C400 (4 per box)
40491E	Limb Plate Electrode C400 (4 per bag)
40424A	Limb Plate Electrode (4 pack)
14030A	Strap Electrode, 14 inch limb (4 per pack)

Printer Paper

Part Number	Description
M3707A	Thermal Paper (100 sheets), A size (8.5 x 11 in/21.6 x 28 cm)
M3708A	Thermal Paper (100 sheets) A4 size (8.27 x 11.69 in/21 x 29.69 cm)

Battery

Part Number	Description
989803130151	PageWriter Trim Replacement Battery

Cardiograph Accessories

For more information on optional cardiograph accessories or to place an order, contact your Philips Sales Representative the Philips Response Center nearest you, see page 1-50.

Optional Cardiograph Accessories

Part Number	Description
989803129931	Barcode Reader
989803129941	Magnetic Card Reader
989803127331	PC card (128 MB storage)
989803129961	SmartCard Reader
989803145331	USB Memory Stick
989803130131	Patient Cable Arm
989803142041	Wireless LAN Card

Optional Cardiograph Accessories

Part Number	Description
989803129951	LAN PCMCIA Network Card
989803144601	PageWriter Trim Cardiograph User Documentation CD (additional copies)
989803130111	PageWriter Trim Cardiograph Cart
989803130121	PageWriter Trim Cardiograph Cart Additional Shelf
989803127441	Optional Locking Drawer for Cardiograph Cart
989803127461	Modem Card (USA and Canada only)
989803138021	LAN Cable

Contacting a Philips Response Center

The Philips Response Center can assist with product troubleshooting and provide technical expertise to help with any issue with the PageWriter Trim cardiograph or any of its accessories.

For more information on the Philips Response Center go to:

www.medical.philips.com/main/services/response_center

North America Response Centers

Country	Telephone Number
Canada	(800) 323 2280
Mexico	01 800 710 8128
Puerto Rico	1 787 754 6811
United States	(800) 722 9377

South America Response Centers

Country	Telephone Number
Argentina	54 11 4546 7698
Brazil	0800 701 7789
Chile	0800 22 3003
Columbia	01 8000 11 10 10
Peru	51 1 620 6440

Europe Response Centers

Country	Telephone Number
United Kingdom	44 0870 532 9741 Fax: 44 01737 23 0550
Austria	43 1 60101 820
Belgium	32 2 525 7102 (French) 32 2 525 7103 (Flemish)
Czech Republic MCR Response Center (located in The Netherlands)	31 40 2781619
Denmark	45 80 30 30 35
Finland	358 615 80 400
France	0 810 835 624
Germany	0180 5 47 5000

Europe Response Centers *(continued)*

Country	Telephone Number
Greece MCR Response Center (located in The Netherlands)	31 40 2781619
Hungary MCR Response Center (located in The Netherlands)	31 40 2781619
Italy	0800 232100
Netherlands	31 40 27 211 27
Norway	47 800 84 080
Poland MCR Response Center (located in The Netherlands)	31 40 2781619
Rumania MCR Response Center (located in The Netherlands)	31 40 2781619
Russia MCR Response Center (located in The Netherlands)	31 40 2781619
Slovak Republic MCR Response Center (located in The Netherlands)	31 40 2781619

Europe Response Centers *(continued)*

Country	Telephone Number
Spain	34 90 230 4050
Sweden	46 200 81 00 10
Switzerland	0800 80 3000 (German) 0800 80 3001 (French)

Asia Response Centers

Country	Telephone Number
Australia	1800 251 400
China	800 810 0038
Hong Kong	852 2876 7578
India New Delhi Mumbai Chennai Bangalore Hyderabad	 011 2695 9734 022 5691 2463/2431 044 5550 1141 080 5692 9911 040 5578 7975
Indonesia	62 21 7910040, ext 8610
Japan	81 0120 095 205
Korea	82 02 3445 9010
Malaysia	1800 886 188
New Zealand	0800 251 400
Philippines	63 2 8162617 ext. 875
Singapore	1800 Philips

Asia Response Centers *(continued)*

Country	Telephone Number
Taiwan	0800 005 616
Thailand	02 614 3569

Africa and Middle East

Country	Telephone Number
All countries MCR Response Center (located in The Netherlands)	31 40 2781619

2

PageWriter Trim II and III Configuration

Introduction

Settings on the PageWriter Trim II and III cardiographs are configurable to meet the needs of any clinical environment. The cardiograph is shipped with factory default settings, and any of these settings may be changed.

There are no configurable features on the PageWriter Trim I cardiograph. Please see the *PageWriter Trim I, II, III Instructions for Use* on the User Documentation CD, and complete the *Interactive Training Program* on CD to learn how to use the features of the cardiograph. These files may also be downloaded from the Philips InCenter web site (incenter.medical.philips.com). For information on using the Philips InCenter web site, see page 1-4.

Configuring Multiple Cardiographs

When configuring multiple cardiographs with the same settings, configure one cardiograph with all custom settings and then save the configured settings as a *configuration settings file* to a PC card or to a USB memory stick. The configuration settings file may then be uploaded to additional cardiographs. See "Saving Configured Settings" on page 2-64 for more information.

Password Access

Access to all configuration options on the cardiograph may be set to be password access controlled. It is recommended that the Password screen be configured first, see page 2-56.

Configuring the Wireless LAN Card

For information on configuring the wireless LAN option for the PageWriter Trim cardiograph, please see the *PageWriter Trim Cardiograph Wireless LAN Installation Instructions* included on the User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com). For information on using the Philips InCenter web site, see page 1-4.

Configuration with a Philips TraceMasterVue ECG Management System

If a Philips TraceMasterVue ECG Management System is used with the PageWriter Trim cardiograph, please note that certain types of ECG information will not be transferred from the cardiograph to TraceMasterVue.

For detailed information on configuring the PageWriter Trim cardiograph with a TraceMasterVue ECG Management System, see the *Configuring Cardiograph and TraceMasterVue Communication Guide* available for download from the Philips InCenter web site (incenter.medical.philips.com). For information on using the Philips InCenter web site, see page 1-4.

Restoring Cardiograph Configured Settings

After configuring the cardiograph, save the settings to a PC card or to a USB memory stick in case of system failure. If the cardiograph loses configured settings, the settings may be quickly and easily restored by loading them onto the cardiograph with a PC card or a USB memory stick. See "Loading the Configuration Settings File" on page 2-66 for more information.

Opening the Configuration Screen

All custom settings for the cardiograph are specified on the Configuration screens.

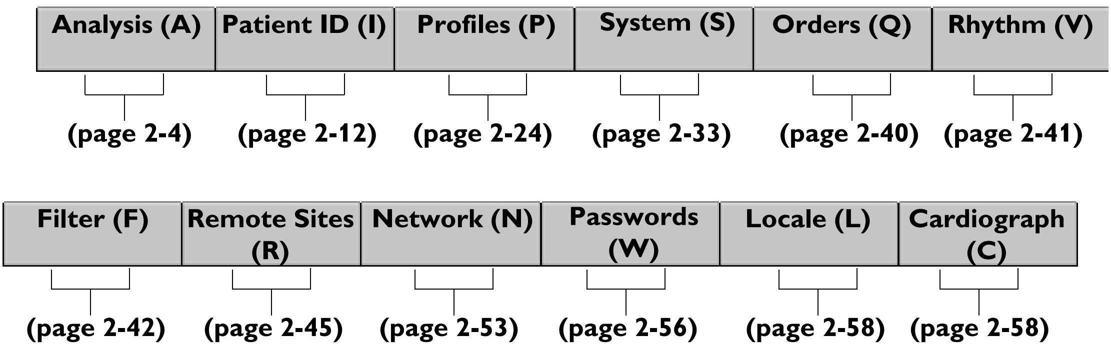
To open the configuration screen:

1 Press the *Tab* key ( ) or turn the Trim Knob to highlight the **Config** button on the Command Toolbar.

2 Press the space bar ( ) or the Trim Knob to select the button. The Configuration screen appears.

A series of tabs appear at the top of the screen. Under each tab are a set of configurable features.

Figure 2-1 The Configuration Screen



To select a tab:

1 Press and hold down the *Alt* key ( ) on the keyboard.

2 Press the letter shown in parenthesis on the tab. For example, to select the Analysis tab, press and hold down the *Alt* key, and then press the *A* key on the keyboard.

To save settings on any configuration screen:

▶ After making all changes to the cardiograph configuration screens, press the *Enter* key ( ) on the keyboard to save the settings and to exit the screen.

To exit the configuration screens without saving changes:

- Press the *Esc* key ( ) on keyboard to exit any configuration screen without saving the information.

Analysis Settings

This section is used to configure the settings used with the Philips 12-Lead Algorithm, including selecting the algorithm version, setting the default pacing detection setting, the adult bradycardia limit, and configuring the statement that appears on the printed 12-lead ECG indicating that the ECG has not been overread by a qualified physician.

About Algorithm Versions

The PageWriter Trim cardiograph supports two versions of the Philips 12-Lead Algorithm. Both versions of the algorithm may be configured to provide interpretive, reason, and severity statements that appear on the ECG report.

The algorithm version **PH080A** is the factory default algorithm, and is compatible with all versions of the TraceMaster and TraceMasterVue ECG Management System, and with the PageWriterTouch cardiograph.

The algorithm version **PH090A** is only compatible with the TraceMasterVue ECG Management System version A.02.01 and higher. This version provides all of the same features as version PH080A, with the following enhancements:

- heightened detection of atrial arrhythmias including: atrial fibrillation, atrial flutter, and second and third degree AV blocks
- ability to suppress interpretive statements that indicate a borderline condition
- optional automatic lead reversal detection feature for both limb leads (RA and LA), and the precordial leads
- calculation of Fridericia rate corrected QT interval

Factory Default Settings

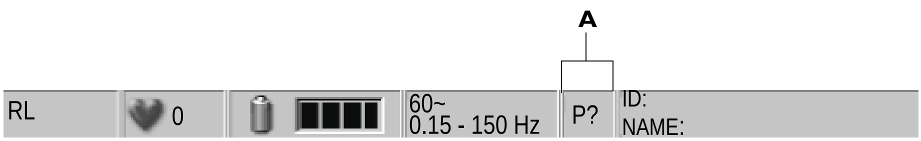
- Philips 12-Lead Algorithm Version: **PH080A**
- Pacing Detection Setting: **Not Known if Paced** (**P?**) (recommended for most ECGs)
- Report Confirmation Label: **Unconfirmed Diagnosis**
- Adult Bradycardia Limit: 50 (bpm)

Setting the Default Pacing Detection Setting

The default pacing detection setting appears on the Status Bar at the beginning of each patient session.

NOTE The default Pacing Detection Setting may be changed at any time during a patient session by selecting the **Report** button on the Command Toolbar (see page 1-45).

Figure 2-2 Pacing Detection Setting on the Status Bar



A Pacing Detection Setting

There are four pacing detection settings that are available as the default setting on the cardiograph. These settings are described in Table 2-1. The pacing detection setting may be changed for an individual patient session when necessary.

Table 2-1 Pacing Detection Setting Description

Setting	Description	Symbol on Status Bar and printed ECG
Not known if paced	■ This is the default setting and normally is used for both paced and non-paced patients ■ Pacemaker pulse detection is on and is at normal sensitivity ■ Occasional false pacemaker pulse detections may occur in ECGs with excessive noise ■ False detections may result in an incorrect interpretive statement appearing on the report ■ Small amplitude pacemaker pulses may not be detected using this setting	**P?**

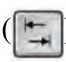
Table 2-1 Pacing Detection Setting Description *(continued)*

Setting	Description	Symbol on Status Bar and printed ECG
Non-paced	■ Pacemaker pulse detection is off ■ Use this setting if there are false pacemaker pulse detections from noise, or if incorrect interpretive statements or inappropriate paced ECG complexes appear on the report	No code appears on the Status Bar if the Non-paced setting is selected
Paced	■ Pacemaker pulse detection is on and is set at a higher sensitivity ■ Use this setting if small amplitude pacemaker pulses are not being detected at the default (**Not Known if Paced**) setting ■ False pacemaker pulse detections may occur if the ECG is noisy	**P**

Table 2-1 Pacing Detection Setting Description *(continued)*

Setting	Description	Symbol on Status Bar and printed ECG
Paced (magnet)	▪ Use this setting if the ECG is acquired with an active pacemaker magnet or programmer in place ▪ Pacemaker pulse detection is on and is at a higher sensitivity ▪ Magnets or programmers often put the pacemaker into a fixed-rate, non-sensing mode ▪ The statement **ECG ACQUIRED WITH MAGNET IN PLACE** is printed on the 12-lead ECG notifying the overreader that a magnet or programmer was used and would explain the fixed rate behavior of the pacer	**PM**

To set the default pacing detection setting:

1. Press and hold down the *Alt* key () on the keyboard.
2. Press the *A* key on the keyboard.
3. The Analysis screen appears. Under **Pacer Detect**, **Not Known if Paced** (recommended setting) is selected.
4. To select another setting, turn the Trim Knob or press the *Tab* key () to highlight the setting.
5. Press the space bar (on keyboard) or the Trim Knob to select the setting.

Setting the Report Status Statement

The statement **Unconfirmed Diagnosis** or any other customized text can be configured to appear on the ECG report to designate that the report has not been overread by a qualified physician. To have the default statement **Unconfirmed Diagnosis** appear on the printed ECG report, do not change this setting. To customize a statement or to have no statement appear on the report, follow the procedure below.

To configure the report status statement:

1. Turn the Trim Knob or press the *Tab* key ( ) to select the field under **Report Confirmation Label**.
2. Press the space bar (on keyboard) to delete the statement. No statement will appear on the printed ECG report. Or, type in a customized statement to appear on the ECG report.

Setting the Default Adult Bradycardia Limit

The default adult bradycardia limit that is used when interpreting waveform data with the Philips 12-lead Algorithm is configurable. For more information on adult bradycardia limits and the Philips 12-Lead Algorithm, see the *Philips 12-Lead Algorithm Physician's Guide* on the PageWriter Trim User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com).

To set the adult bradycardia limit setting:

1. Press the *Tab* key ( ) or turn the Trim Knob to select the setting under **Adult Bradycardia Limit**.
2. Press the space bar (on keyboard) or the Trim Knob. The drop-down list of available settings appears.
3. Press the up or down arrow keys ( ) or turn the Trim Knob to highlight a setting.
4. Press the space bar (on keyboard) or the Trim Knob to select the setting.

Setting the Philips 12-Lead Algorithm Version

The default algorithm version **PH080A** may be changed to algorithm version **PH090A**. For information on algorithm version **PH090A**, see “About Algorithm Versions” on page 2-4.

To select the algorithm version:

1. Press the *Tab* key ( ) or turn the Trim Knob until the algorithm version (**PH080A** or **PH090A**) is highlighted.
2. Press the space bar or the Trim Knob to select the algorithm version.
3. If version **PH090A** is selected, a group of configurable options appear on the screen.

NOTE Algorithm version PH090A is only compatible with TraceMasterVue ECG Management System version A.02.01 and higher, and is not compatible with the PageWriterTouch cardiograph.

4. Press the *Tab* key or turn the Trim Knob to highlight the **Check Lead Placement** setting. Press the space bar to select the check box. This option enables the automatic lead reversal detection feature for limb leads RA and LA, and the precordial leads. When the algorithm detects that leads have been reversed, a statement will appear on the ECG report.

About the Borderline Statement Suppression Setting

This feature may be configured to suppress interpretive statements that indicate a *borderline* or *otherwise normal* condition. These borderline interpretive statements are generated by measurements that are above an abnormal threshold, but may in fact indicate a non-pathological condition. These statements indicate to the clinician that a condition may be present, but there is no decisive indicator. Borderline interpretive statements include the terms *minimal*, *consider*, *borderline*, or *probable*.

The borderline statement suppression setting is configurable to three levels of sensitivity as described in Table 2-2.

For a complete listing of all borderline interpretive statements that are suppressed using this feature, see Appendix A, “Suppressed Borderline Interpretive Statements.”

Table 2-2 Borderline Statement Suppression Settings Description

Borderline Suppression Setting	Description
Level 0 - none	■ This setting turns off all borderline statement suppression; no statements are suppressed. ■ All *borderline* or *otherwise normal* interpretive statements will appear on the printed ECG report.
Level 1 - few	■ This setting will suppress certain borderline interpretive statements. ■ See Appendix A, “Suppressed Borderline Interpretive Statements” for a complete listing of the interpretive statements that are suppressed using this setting.
Level 2 - many	■ This setting will suppress the maximum number of borderline or otherwise normal interpretive statements. ■ See Appendix A, “Suppressed Borderline Interpretive Statements” for a complete listing of the interpretive statements that are suppressed using this setting.

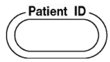
To select a borderline statement suppression setting:

1 Press the *Tab* key ( ) or turn the Trim Knob until the setting is highlighted.

2 Press the space bar to select the radio button.

Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key ( ) to save the changes and to exit the Configuration screen. The new changes are applied.

Patient ID Settings



The Patient ID settings are used to configure the amount and type of information that appears on the Patient ID screen when the **Patient ID** button (right side of the cardiograph) is pressed. The Patient ID screen is used during a patient session to enter patient information.

CAUTION **Age** and **Gender** Patient ID information are required if using the Philips 12-Lead Algorithm for ECG interpretation. For more information, see the *Philips 12-Lead Algorithm Physician's Guide* on the PageWriter Trim User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com).

To open the Patient ID configuration screen:

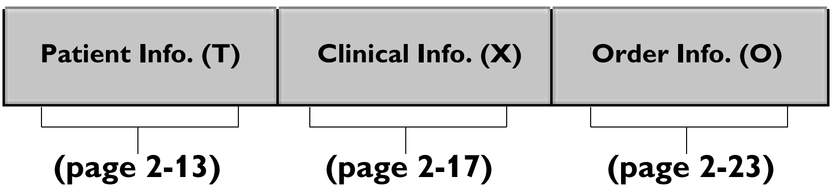
1 Push and hold down the *Alt* key (

) on the keyboard.

2 Press the *I* key on the keyboard to open the Patient ID configuration screen.

3 Three tabs appear at the top of the screen. These tabs contain the three types of configurable Patient ID information available on the cardiograph. The **Patient Info.** tab is selected.

Figure 2-3 Patient ID Configurable Settings



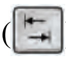
Patient ID Configuration Options

Each type of Patient ID information may be configured with up to four options as described in Table 2-3. These options appear as columns of check boxes on the left side of the screen. Select the check box underneath the option to enable it.

Table 2-3 Patient ID configuration options

Setting	Description
Enabled	■ Select this check box to make this type of Patient ID information appear on the Patient ID screen. ■ This information is transferred with the ECG to a TraceMasterVue ECG Management System.
Required	Select this check box to require that this type of Patient ID information be complete before the ECG is transferred to a Philips TraceMasterVue ECG Management System.
Indexed	Select this check box to make this type of Patient ID information appear as a column on the Archive or Orders screen.
Printed	Select this check box to include this type of Patient ID information on the printed ECG report.

To select a check box:

1 Turn the Trim Knob or press the *Tab* key () to highlight the **ON** text next to the check box.

2 Press the space bar (on keyboard) or the Trim Knob to select the check box. A check mark appears and the feature is enabled.

Configuring Patient Information

These selections include demographic information about the patient including: name, ethnicity, age, gender, and other configurable information fields. This information appears under the **Patient Info.** tab when the **Patient ID** button (right side of cardiograph) is pressed.

There are two optional and configurable text entry fields available labeled **<User Field 1>** and **<User Field 2>**. These optional text entry fields may be configured to appear on the Patient ID screen during a patient session.

Each type of Patient Information can be configured using four different settings: **Enabled**, **Required**, **Indexed,** and **Printed** (see Table 2-3 on page 2-13).

Table 2-4 Patient Information Description

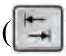
Patient Information	Description
Patient First Name	Up to 40 characters in length
Patient Last Name	Up to 40 characters in length
Patient Additional Name	Up to 40 characters in length
Race (Ethnicity)	■ Drop-down list selection ■ Pre-defined drop-down list
Patient Age	■ Patient age is configurable and can be entered by: – Configurable units (years, hours, days, weeks, months) – Date of birth (month, day, year) ■ A default age is specified for ECG interpretation when no patient age is entered
Patient Gender	Radio button selection (male, female, unknown)
User Defined text entry field	■ Text entry field label up to sixteen characters in length ■ Text entry field is free text entry up to thirty-two characters in length

Factory Default Settings

Patient ID fields selected:

- Patient last name (Enabled, Printed)
- Age and date of birth (Enabled, Printed)
- 50 (default patient age used when no age is entered)
- Gender (Enabled, Printed)

To configure Patient Information:

1. Press and hold down the *Alt* key () on the keyboard.
2. Press the *I* key. The Patient ID screen appears.
3. Press and hold down the *Alt* key on the keyboard.
4. Press the *T* key to select the **Patient Info.** tab (if not selected).
5. Press the *Tab* key () or turn the Trim Knob to highlight the **ON** text next to a check box of a feature to be enabled.
6. Press the space bar (on keyboard) or the Trim Knob to select a check box. A check mark appears in the selected check box and enables the patient information field.
7. Press the *Tab* key or turn the Trim Knob to select the drop-down list next to **Age/DOB**.
8. Press the space bar (on keyboard) or the Trim Knob to display the drop-down list.
9. Press the up or down arrow keys to scroll through the list and to highlight a default age unit (**DOB**-date of birth, **Years**, **Days**, **Weeks**, **Hours**).
10. Press the space bar or the Trim Knob to select the age unit.
11. Press the *Tab* key or turn the Trim Knob to highlight the age displayed under **Default Age (Years)**. The **Default Age (Years)** selection sets the

default age that is used for ECG interpretation when no patient age is entered. The factory default age setting is 50 years.

12 Type in the default age (using keyboard).

13 Press the *Tab* key or turn the Trim Knob to highlight the drop-down list next to **Weight** or **Height**.

14 Press the space bar or the Trim Knob to display the drop-down list.

15 Press the up or down arrow key (↑ ↓) to highlight a weight or height unit.

16 Press the space bar or the Trim Knob to select the weight or height unit.

User Defined Patient Information

There are two optional Patient Information text entry fields labeled **<User Field 1>** and **<User Field 2>**. When the **Patient ID** button is pressed during a patient session, these configurable text entry fields appear under the **Patient Info.** tab, and are located at the bottom of the screen.

A text label is created in Configuration to describe the text entry field. The text entry field is completed on the Patient ID screen during a patient session by typing information in the field.

To configure the text field label:

1 Ensure that **<User Field 1>** and **<User Field 2>** check boxes for **Enabled**, **Required**, **Indexed**, or **Printed** are selected. If these check boxes are not selected, the user defined fields will not appear on the Patient ID screen.

2 Turn the Trim Knob or press the *Tab* key ( ) to select the field next to **<User Field 1>** or **<User Field 2>**. The cursor appears in the field.

3 Type in any sixteen letters or numbers with the keyboard to create the text field label. This label appears on the Patient ID screen.

Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key ( ) to save the changes and to exit the Configuration screen. The new changes are applied.

Configuring Clinical Information

The following Clinical Information fields may be configured to appear on the Patient ID screen:

- Symptoms (Sx)
- History (Hx)
- Prescriptions (Rx)
- Diagnoses (Dx)
- Diagnostic Related Group (DRG)
- Blood Pressure

On the Patient ID screen, the user selects Clinical Information from a pre-defined list of entries, or can manually enter information in the clinical fields.

There are two optional and configurable text entry fields available labeled **<User Field 3>** and **<User Field 4>**.

Each type of Clinical Information can be configured using four different settings: **Enabled**, **Required**, **Indexed,** and **Printed** (see Table 2-3 on page 2-13).

Factory Default Settings

- No Clinical Information fields are selected

Table 2-5 Clinical Information Description

Clinical Information	Description
Sx (Symptoms)	■ Configurable drop-down list ■ Selections from drop-down list can be up to thirty-two characters in length

Table 2-5 Clinical Information Description *(continued)*

Clinical Information	Description
Hx (History)	■ Configurable drop-down list ■ Selections from drop-down list can be up to thirty-two characters in length
Rx (Prescriptions)	■ Configurable drop-down list ■ Selections from drop-down list can be up to thirty-two characters in length
Dx (Diagnoses)	■ Configurable drop-down list ■ Selections from drop-down list can be up to thirty-two characters in length
DRG (Diagnostic Related Group)	■ Configurable drop-down list ■ Selections from drop-down list can be up to thirty-two characters in length
Blood Pressure	■ Numeric entry ■ Systolic, diastolic
User Defined text entry field	■ Drop-down list label up to sixteen characters in length ■ Text field is free text entry up to thirty-two characters in length

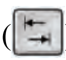
To configure Clinical Information:

1 Press and hold down the *Alt* key () on the keyboard.

2 Press the *I* key. The Patient ID screens appear.

3 Press and hold down the *Alt* key on the keyboard.

4 Press the *X* key to select the **Clinical Info.** tab.

5 Press the *Tab* key () or turn the Trim Knob to highlight the **ON** text next to a check box of a feature to be enabled.

6 Press the space bar (on keyboard) or the Trim Knob to select a check box. A check mark (☑) appears in the selected check box and enables the clinical information field.

7 Press the *Tab* key or turn the Trim Knob to highlight the drop-down list next to **Sx** (Symptoms).

8 Press the space bar or the Trim Knob to review the available selections in the drop-down list.

9 To add a new entry, press the *Tab* key or turn the Trim Knob to highlight the **+** button.

10 Press the space bar or the Trim Knob. The **Please input data** window appears.

11 Type in the new symptom (up to thirty-two characters in length).

12 Press the *Tab* key or turn the Trim Knob to highlight the **OK** button.

13 Press the space bar or Trim Knob to add the symptom. The new symptom appears in the drop-down list.

Follow the same procedure to add drop-down list entries for **Hx** (History), **Rx** (Prescriptions), **Dx** (Diagnosis), and **DRG** (Diagnostic Related Group)

To delete a user-defined symptom:

1 Select the symptom to be deleted from the drop-down list.

NOTE Factory default symptoms cannot be deleted from the drop-down list.

2 Press the *Tab* key or turn the Trim Knob to highlight the **-** button.

3 Press the space bar or the Trim Knob. The **Delete Sx Label** window appears.

4 Press the space bar or the Trim Knob to select the **OK** button. The user-defined symptom is deleted from the drop-down list.

Follow the same procedure to delete drop-down list entries for **Hx** (History), **Rx** (Prescriptions), **Dx** (Diagnosis), and **DRG** (Diagnostic Related Group).

User Defined Clinical Information

There are two optional Clinical Information text entry fields that can be configured labeled **<User Field 3>** and **<User Field 4>**. When the **Patient ID** button is pressed during a patient session, these configurable text entry fields appear under the **Clinical Info.** tab, and are located at the bottom of the screen.

A text label is created in Configuration to describe the text entry field. The text entry field is completed on the Patient ID screen during a patient session by typing information in the field.

To configure the text label:

1. Ensure that the **<User Field 3>** and **<User Field 4>** check boxes for **Enabled**, **Required**, **Indexed**, or **Printed** are selected. If these check boxes are not selected, the user defined fields will not appear on the Patient ID screen or on the printed ECG.
2. Turn the Trim Knob to select the field next to **<User Field 3>** or **<User Field 4>**. The cursor appears in the field.
3. Type in any sixteen letters or numbers with the keyboard to specify a text label for the field.

   Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key ( ) to save the changes and to exit the Configuration screen. The new changes are applied.

Configuring Order Information

The following Order information may be configured to appear on the Patient ID screen:

- Reason for Order
- Physician Name
- UPIN (Universal Physician Identification Number)
- Department
- Operator ID
- Room number
- Order Number

- Encounter ID
- STAT ECG

There are two optional and configurable text entry fields available labeled **<User Field 5>** and **<User Field 6>**. These optional text entry fields appear on the Patient ID screen during a patient session.

Each type of Order Information can be configured using four different settings: **Enabled**, **Required**, **Indexed,** and **Printed** (see Table 2-3 on page 2-13).

Factory Default Settings

- No Order Information fields are selected

Table 2-6 Order Information Description

Order Information	Description
Reason for Order	■ Configurable drop-down list ■ Drop-down list selections up to thirty-two characters in length
Physician Name and UPIN (Universal Physician Identification Number)	■ Configurable drop-down list ■ Drop-down list selections up to thirty-two characters in length
Department Number	■ Configurable drop-down list ■ Drop-down list selections up to thirty-two characters in length

Table 2-6 Order Information Description *(continued)*

Order Information	Description
Operator ID	■ Configurable drop-down list ■ Drop-down list selections up to thirty-two characters in length
Room Number	■ Free text entry ■ Up to thirty-two characters in length
Order Number	■ Free text entry ■ Up to thirty-two characters in length
Encounter ID	■ Free text entry ■ Up to thirty-two characters in length
STAT ECG	■ Selectable drop-down list on Patient ID screen ■ STAT label printed on Auto ECG
User Defined text entry fields	■ Text entry field label up to sixteen characters in length ■ Free text entry up to thirty-two characters in length

To configure Order Information:

1. Press and hold down the *Alt* key (Alt) on the keyboard.
2. Press the *I* key. The Patient ID screens appear.
3. Press and hold down the *Alt* key on the keyboard.
4. Press the *O* key to select the **Order Info.** tab.
5. Press the *Tab* key ( ) or turn the Trim Knob to highlight the **ON** text next to a check box of a feature to be enabled.
6. Press the space bar or the Trim Knob to select a check box. A check mark ( ) appears in the selected check box and enables the Order information field.

7 Press the *Tab* key or turn the Trim Knob to highlight the **+** button next to **Reason for Order**.

8 Press the space bar (on keyboard) or the Trim Knob. The **Please input data** window appears.

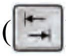
9 Type in a reason for order (up to thirty-two characters in length).

10 Press the *Tab* key or turn the Trim Knob to highlight the **OK** button.

11 Press the space bar or the Trim Knob to add the reason for order. The new reason for order appears in the drop-down list.

To delete a reason for order from the drop down list:

1 Select the reason for order to be deleted from the drop-down list.

2 Press the *Tab* key () or turn the Trim Knob to highlight the **-** button.

3 Press the space bar (on keyboard) or the Trim Knob. The **Delete Reasons** window appears.

4 Press the space bar or the Trim Knob to select the **OK** button. The reason for order is deleted from the drop-down list.

Follow the same procedure to add or to delete drop-down list entries for **Physician Name** and **UPIN, Department,** and **Operator ID**.

User Defined Order Information

There are two optional Order Information text entry fields that can be configured labeled **<User Field 5>** and **<User Field 6>**. When the Patient ID button is pressed during a patient session, these configurable text entry fields appear under the **Order Info.** tab on the Patient ID screen.

A text label is created in Configuration to describe the text entry field. The text entry field is completed on the Patient ID screen during a patient session.

To configure the text label:

1 Ensure that the **<User Field 5>** and **<User Field 6>** check boxes for **Enabled**, **Required**, **Indexed**, or **Printed** are selected. If these check

boxes are not selected, the user defined fields will not appear on the Patient ID screen or on the printed ECG.

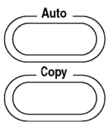
2. Turn the Trim Knob or press the *Tab* key ( ) to select the field next to **<User Field 5>** or **<User Field 6>**. The cursor appears in the field.

3. Type in any sixteen letters or numbers with the keyboard. The text field label is entered and will appear on the Patient ID screen.

   Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key ( ) to save the changes and to exit the Configuration screen. The new changes are applied.

Configuring Auto ECG Report Profiles

TIP Review Chapter 5 “Understanding the Printed Report” of the *PageWriter Trim Cardiograph Instructions for Use* before configuring Auto ECG report profiles. This file is available on the User Documentation CD, or may be downloaded from the Philips InCenter web site (incenter.medical.philips.com).



A *profile* is a group of settings applied to Auto ECGs and to copies of Auto ECGs. A profile defines how the Auto ECG or ECG copy appears when printed. The Auto ECG or ECG copy is generated when the **Auto** or **Copy** buttons (right side of cardiograph) are pressed during a patient session.

Figure 2-4 The Report and Copy Profile

The Profile

- Speed (mm/sec) and Size (mm/mV) of waveforms
- Lead Sequence: Standard or Cabrera
- ECG Signal Acquisition: Time-Sequential or Simultaneous
- Lead Configuration on printed report (3x4, 3x4 1R, 3x4 3R, 6x2, 12x1, Pan-12)
- Extended Measurements Report: enabled or not enabled
- Level of ECG interpretation using the Philips 12-Lead Algorithm (Measurements, Interpretive and Reason Statements, Severity Codes)

There is one factory default report profile (**Default Main**) and one factory default copy report profile (**Default Copy**).

The factory default report and copy profile may be edited, or new, customized profiles may be created.

NOTE The factory default report profile (**Default Main**) and copy profile (**Default Copy**) may not be deleted.

Figure 2-5 The Factory Default Report Profile (Default Main)

Default Main

- Speed: 25 mm/sec
- Size: 10 mm/mV
- Chest Leads: Full Scale
- Lead Configuration: 3x4, 1R
- Lead Sequence: Standard
- Acquisition: Time-Sequential
- Interpretation Level: None

Figure 2-6 The Factory Default Copy Profile (Default Copy)

Default Copy

- Speed: 25 mm/sec
- Size: 10 mm/mV
- Chest Leads: Full Scale
- Lead Configuration: 3x4, 1R (Rhythm Lead: II)
- Lead Sequence: Standard
- Acquisition: Time-Sequential
- Interpretation Level: None
- Number of report copies printed: 1

To open the profile screens:

1 Press and hold down the *Alt* key ( ) on the keyboard.

2 Press the *P* Key on the keyboard. The Profile screen appears.

Under **Profiles List** (left side of screen) **Default Main** is highlighted. Default Main is the factory default Auto 12-lead ECG report profile. This profile may be edited or new profiles may be created.

Four tabs appear on the left side of the Profiles screen. Under each tab are ECG report settings that may be edited.

Figure 2-7 The Profile tabs

Format (M)	Interps (G)	Scale (J)	Type (K)
(page 2-27)	(page 2-29)	(page 2-29)	(page 2-29)

NOTE The **Interps (G)** and **Type (K)** tabs only appear if the Philips 12-Lead Algorithm is enabled on the cardiograph.

To edit a factory default or existing profile:

1 Under **Profiles List** (left side of screen) an Auto ECG report profile is highlighted. This is the profile selected to be edited. The settings for this profile are displayed on the screen. To select a different profile for editing, press the space bar or the Trim Knob to display the drop-down list.

2 Press the up or down arrow key ( ) to scroll through the list and to highlight a profile to be edited.

3 Press the space bar (on keyboard) or the Trim Knob to select a the report profile for editing. The drop-down list closes and the selected report profile appears highlighted under **Profiles List.**

Editing the Profile Report Format

The profile report format settings define the lead sequence and acquisition type that appear on the Auto ECG.

To edit the report format:

1. If the **Format (M)** tab is not selected, press and hold down the *Alt* key () on keyboard. Press the *M* key. The Format screen appears.
2. Press the *Tab* key () or turn the Trim Knob until the drop-down list under **Lead Configuration** is highlighted.
3. Press the space bar (on keyboard) or the Trim Knob to display the list of available report formats. Press the up or down arrow keys () to scroll through the list.
4. Press the space bar or the Trim Knob to select a report format. The selected report format appears highlighted under **Lead Configuration**.

   If the selected report format includes rhythm strips, the default rhythm leads are displayed under **Rhythm**.
5. Press the *Tab* key or turn the Trim Knob to highlight the **Lead Groups>>>** button (3x4, 3R only). Press the space bar or the Trim Knob to display pre-defined groups of three rhythm leads. Or, turn the Trim Knob or press the *Tab* key to highlight the drop-down list next to **R1** (**R2**, **R3**) and select the rhythm leads individually.
6. Under **Lead Sequence**, the default settings for the selected report format are displayed. Press the *Tab* key or turn the Trim Knob to highlight the lead sequence drop-down list (**Standard**, **Cabrera**).

NOTE The Pan-12 report format is only available with **Cabrera** lead sequence and **Time-Sequential** acquisition.

NOTE The 12x1 report format is only available with **Simultaneous** acquisition.

7. Press the space bar or the Trim Knob to display the drop-down list.
8. Press the up or down arrow key to highlight a lead sequence.
9. Press the space bar to select a lead sequence.
10. Follow the same procedure to select an ECG acquisition type (**Simultaneous**, **Time-Sequential**) for the report profile.

Editing Profile Interpretation

The interpretation settings define how much interpretive information from the Philips 12-lead Algorithm appears on the Auto ECG.

To edit interpretation options:

1 Press and hold down the *Alt* key ( ).

2 Press the *G* key on the keyboard. The Interpretation screen appears.

Under **Interpretations,** the different levels of interpretive information available with the Philips 12-lead Algorithm are displayed.

3 Press the *Tab* key ( ) or turn the Trim Knob to highlight the desired setting.

NOTE Selecting **None** will have no interpretive information (including Basic Measurements) appear on the printed Auto ECG.

4 Press the space bar (on keyboard) or the Trim Knob to select the interpretive setting.

Editing Profile Scale Information

The scale settings define the speed and the size of the waveforms appearing on the printed report.

1 Press and hold down the *Alt* key ( ) on the keyboard.

2 Press the *J* key. The Scale screen appears.

3 Press the *Tab* key ( ) or turn the Trim Knob to select settings under **Speed**, **Size**, and **Chest Leads**.

NOTE **Chest Leads** sets the precordial leads to full or half the scale of the limb and augmented leads.

4 Press the space bar (on keyboard) or the Trim Knob to select a setting.

Editing Report Type Information

The Type setting enables the optional Extended Measurements Report. The Extended Measurements report summarizes the output of the Philips 12-Lead Algorithm and includes the morphology characteristics for the individual leads, along with the rhythm characteristics for the rhythm groups. The algorithm uses this measurement information to generate interpretive statements. The Extended Measurements report is especially useful if you want to examine the measurements used to generate an interpretation.

For more information on the Extended Measurements Report, see the *Philips 12-Lead Algorithm Physician's Guide* on the User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com).

1 Press and hold down the *Alt* key ( ) on the keyboard.

2 Press the *K* key. The Type screen appears.

3 Press the *Tab* key ( ) or turn the Trim Knob to highlight **Extended Measurements**.

4 Press the space bar (on keyboard) to select the check box. With the check box selected, the full Extended Measurements report is included with each Auto ECG.

Saving the Report Profile

The selected settings may be saved to the current report profile, or a new report profile may be created using the selected settings.

To save the settings to the selected report profile:

1 Press the *Tab* key ( ) or turn the Trim Knob until the **Save** button (top of screen) is highlighted. The new settings will be saved to the profile displayed under **Profiles List**.

NOTE To select a different profile, press the *Tab* key or turn the Trim Knob until the drop-down list under **Profiles List** is highlighted. Press the space bar or the Trim Knob to open the drop-down list. Press the up or down arrow key to highlight a profile. Press the space bar or the Trim Knob to select the profile.

2 Press the space bar or the Trim Knob to apply the new settings to the selected profile.

The new settings are saved to the selected profile.

To create a new report profile:

1. Press the *Tab* key ( ) or turn the Trim Knob until the **Add** button (top of screen) is highlighted. Press the space bar or the Trim Knob.
2. The **Please input data** window appears. Type in the name of the new profile (up to thirty-two characters in length).
3. Press the *Tab* key to highlight the **OK** button.

NOTE Highlight the **Cancel** button and push the space bar or Trim Knob to exit without saving the profile name.

4. Press the space bar (on keyboard) or the Trim Knob to save the new profile.

   The new profile appears in the **Profiles List** drop-down list.

Setting the Default Auto Report and Copy Profile

The default Auto report and copy profile are selected on the Profile configuration screen. Any Auto ECG or ECG copy printed on the cardiograph will apply the settings specified in the selected default profile.

During a patient session, the default profile may be changed and temporary settings applied by selecting the **Report** button on the Command Toolbar (see page 1-45).

To set the default report and copy profile:

1. Press the *Tab* key ( ) or turn the Trim Knob until the drop-down list under **Select Defaults (Main)** (right side of screen) is highlighted.
2. Press the space bar (on keyboard) or the Trim Knob to display the drop-down list of available Auto ECG report profiles.
3. Press the up or down arrow key ( ) to highlight a profile as the default Auto ECG report profile.
4. Press the space bar to select the profile.
5. Press the *Tab* key or turn the Trim Knob to highlight the drop-down list under **Copy**.

6 Press the space bar or the Trim Knob to display the drop-down list of available Copy profiles.

7 Press the up or down arrow key to highlight a profile.

8 Press the space bar or the Trim Knob to select the profile as the default Copy profile.

9 Press the *Tab* key or turn the Trim Knob to highlight the drop-down list next to **Number of Copies**.

10 Press the space bar or the Trim Knob to display the drop-down list.

11 Press the up or down arrow key to select the number of Auto ECG report copies (1-5) printed each time the **Copy** button (right side of cardiograph) is pressed.

Setting Other Print Options

Additional print options are available on the Profiles configuration screen. The **Auto Save When Print** option automatically saves all Auto ECGs to the Archive. The Archive serves as temporary storage for ECGs until they are saved to a PC card or to a USB memory stick, or are transferred by modem, fax, or a LAN connection to a TraceMasterVue ECG Management System. ECGs that are not saved to the Archive are deleted.

The **Second ID Label** option prints the patient name and ID at the top and the bottom of the ECG report.

The **Print Immediately** option automatically prints each ECG when the **Auto** button (right side of cardiograph) is pressed. If this option is not selected, the Auto ECG appears on the Preview screen. The user can assess the quality of the ECG directly on the Preview screen, and then print it by pressing the *Enter* key ( ) on the keyboard.

TIP Previewing the Auto ECG before printing it saves printer paper and helps to ensure a quality report.

To select other print options:

1. Press the *Tab* key ( ) or turn the Trim Knob until the option is highlighted (lower right of screen)
2. Press the space bar (on keyboard) to select the check box. A selected check box enables the feature.

   Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key ( ) to save the changes and to exit the Configuration screen. The new changes are applied.

System Settings

The configurable cardiograph settings in this section include:

- paper size
- keyboard caps lock feature
- lead standard setting for printed ECGs (AAMI or IEC)
- battery power saving modes
- default display format
- notification messages
- simulated waveform feature

This section also includes tools under **Configuration Management** (see page 2-64) and other maintenance and diagnostic tools under **Diagnostics**. For information on Diagnostic tools, see Chapter 7 “Care and Maintenance” of the *PageWriter Trim Cardiograph Instructions for Use* on the User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com).

The **Software Option Upgrade** feature is used when installing a software or an accessory upgrade on the cardiograph. For information, see the *PageWriter Trim Cardiograph Upgrade Instructions* provided with the upgrade kit.

Factory Default Settings

- Paper Size (varies by region)
- Caps Lock: Off
- Lead standard (varies by region)
- Default Display Format: 6x2
- Battery Power Saving Modes:
  - Standby: On with Idle Timeout: 5 minutes
  - Shut Down: On with Idle Timeout: 10 minutes
- Notification message: none
- Simulated Data: Not enabled

Setting the Default Paper Size

The cardiograph is shipped configured for the common paper size for your country or region. The paper size can be changed in Configuration.

To change the default paper size:

1. Press down and hold the *Alt* key ( ) on the keyboard.
2. Press the *S* key. The System screen appears.
3. Press the *Tab* key ( ) or turn the Trim Knob until the **A** or **A4** setting under **Paper Size** is highlighted.

NOTE A4 paper requires a special paper drawer modification. Contact the nearest Philips Response Center for more information, see page 1-50.

4. Press the space bar (on keyboard) or the Trim Knob to select a default paper setting.

Setting the Caps Lock Feature

The cardiograph keyboard can be configured to type in all capital letters. When the feature is enabled, the icon A appears on the Order entry and Patient ID screens.

To enable the caps lock feature:

1 Press the *Tab* key ( ) or turn the Trim Knob until **ON** is highlighted under **Caps Lock**.

2 Press the space bar (on keyboard) to select the check box. With the check box selected ( ), the keyboard will type in all capital letters.

Printed ECG Lead Standard

The cardiograph can be configured to print ECGs with the AAMI or IEC lead standard. The R/T (real-time) ECG screen on the cardiograph will always display AAMI leads, regardless of the printed ECG setting.

To set the printed report lead standard:

1 Press the *Tab* key ( ) or turn the Trim Knob until the **AAMI** or **IEC** lead standard is highlighted.

2 Press the space bar (on keyboard) or the Trim Knob to set the default print lead standard.

Default Display Format Setting

The R/T (real-time) ECG screen can be configured to display a default 6x2 or 3x4, 1R lead configuration. Each time the cardiograph is shut down and restarted, the default display format appears.

NOTE The R/T ECG screen displays AAMI lead configuration in the 6x2 or 3x4, 1R format.

The lead format on the R/T ECG screen can be changed at any time during a patient session by pressing the up or down arrow key.

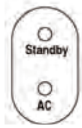
For more information on using the R/T ECG screen, see page 1-40, or see Chapter 3 “The Patient Session” of the *PageWriter Trim Cardiograph Instructions for Use* on the User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com).

To set the default display format setting:

1. Press the *Tab* key ( ) or turn the Trim Knob until the **6x2** or **3x4, 1R** lead format is highlighted.
2. Press the space bar (on keyboard) to select the setting.

Battery Power Saving Modes

The **Battery Power Saving Modes** are used to specify the **Standby** and **Shutdown** features. Both features are fully configurable and are used to help prolong battery life between charges.



The **Standby** feature turns off the cardiograph display (screen is black) if the cardiograph is not used for a pre-set number of minutes. When the cardiograph is in Standby, the green Standby indicator light illuminates (left side of cardiograph). Standby can be entered on any cardiograph screen by pressing the On/Standby button. Press any key on the cardiograph to exit Standby. The cardiograph quickly returns to the previously viewed screen.

NOTE The cardiograph will not enter Standby if the Patient ID screen is open, or if the cardiograph is printing an ECG.

The **Shutdown** feature is used to automatically shut down all cardiograph power if the cardiograph is not used for a pre-set number of minutes after entering Standby. The green Standby indicator light is not lit when the cardiograph is shut down. Press the On/Standby button to turn on the cardiograph. The cardiograph will restart and display a series of information screens before the R/T ECG screen appears.

NOTE The Shutdown feature may be disabled.

To set the battery power saving features:

1. Press the *Tab* key ( ) or turn the Trim Knob until the **Idle Timeout (Minutes)** field next to **Standby** is highlighted.
2. Enter the number of minutes (5-60) that elapse before the cardiograph enters Standby.
3. Press the *Tab* key or turn the Trim Knob until the **Idle Timeout (Minutes)** field next to **Shutdown** is highlighted.

NOTE To disable the Shutdown feature, press the *Tab* key or turn the Trim Knob until **Shutdown** is highlighted. Press the space bar (on keyboard) to clear the check box. The Shutdown feature is disabled.

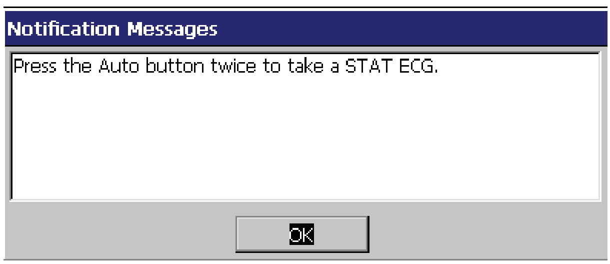
4 Enter the number of minutes (10-3000) that pass after the cardiograph enters Standby that it automatically shuts down.

Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key ( ) to save the changes and to exit the Configuration screen. The new changes are applied.

Setting the Notification Message Feature

The cardiograph can be configured to display a customized message, instruction set, or other information each time that the cardiograph is returned to active use from Standby. The message can contain up to five individual lines of text, with each line up to fifty characters in length. The message displays on the R/T ECG screen until the user selects the **OK** button and closes the window.

Figure 2-8 Sample Notification Message



To create a notification message:

1 Press the *Tab* key ( ) or turn the Trim Knob until the **Add** button under **Notification Messages** is highlighted.

2 Press the space bar (on keyboard) or the Trim Knob to select the button.

3 The **Please input data** window appears. Type in the first line of text to appear on the notification message screen. The information can be up to fifty characters in length.

4 Press the *Tab* key or turn the Trim Knob until the **OK** button is highlighted.

5 Press the space bar or the Trim Knob to enter the line of text into the notification message.

6 Continue to add up to five total lines of text in the order that they should appear on the notification message window.

To delete a line of the notification message:

1 Press the *Tab* key ( ) or turn the Trim Knob until the drop-down list under **Notification Messages** is highlighted (in blue).

2 Press the space bar (on keyboard) or the Trim Knob to open the drop-down list. Press the up or down arrow key to scroll through the lines of text. Highlight the line of text to be deleted.

3 Press the space bar or the Trim Knob to close the drop-down menu.

4 Press the *Tab* key or turn the Trim Knob to highlight the **Delete** button.

NOTE The line of text is automatically deleted once the **Delete** button is selected. No confirmation message to delete is provided.

5 Press the space bar or the Trim Knob to delete the selected line of text from the notification message.

Simulated Waveform Data Feature

The cardiograph can be configured to display simulated waveforms (no data is acquired from the Patient Interface Module). This simulated waveform display may be helpful when training users on cardiograph operation.

When simulated waveforms appear on the screen, the statement **Simulated Data** appears on the R/T ECG screen in the same location as the Patient ID number, see “PageWriter Trim II and III Status Bar” on page 1-44. No Patient ID information can be entered when simulated waveforms are displayed on the cardiograph.

Any ECG printed using simulated waveforms will include the statement **Simulated data** on the report in the same location as the Patient ID number.

To turn on simulated waveform data:

1. From any configuration screen, press down and hold the *Alt* key ( ) on the keyboard.
2. Press the *S* key. The System screen appears.
3. Press the *Tab* key ( ) or turn the Trim Knob until the **ON** check box under **Simulated Data** is highlighted.
4. Press the space bar (on keyboard) or the Trim Knob to select the check box. The simulated waveform feature is enabled.
5. To use the simulated waveform data feature immediately, press the *Enter* key ( ) to save the new configured setting.
6. Press and hold the On/Standby button for three seconds and then release it to shut down the cardiograph.
7. Remove the PIM from the PIM connector port on the rear of the cardiograph (see page 1-15).
8. Press the On/Standby button to restart the cardiograph. Simulated waveforms appears on the R/T ECG screen.

To turn off simulated waveform data:

1. From any configuration screen, press down and hold the *Alt* key ( ) on the keyboard.
2. Press the *S* key. The System screen appears.
3. Press the *Tab* key ( ) or turn the Trim Knob until the **ON** check box under **Simulated Data** is highlighted.
4. Press the space bar (on keyboard) or the Trim Knob to clear the check box. The simulated waveform feature is disabled.
5. Press the *Enter* key ( ) to save the new configured setting.
6. Press and hold the On/Standby button for three seconds and then release it to shut down the cardiograph.
7. Attach the PIM to the PIM connector port on the rear of the cardiograph (see page 1-15).

8 Press the On/Standby button to restart the cardiograph. The cardiograph returns to normal operation.

Order Settings

The Order settings are used with the Orders feature on the cardiograph. The **Quick Search Fields** feature is used on the Patient ID entry screen. This feature is used to automatically locate a Pending Order from the Pending Orders list based on information entered into one searchable field. The field that is used to search for Pending Orders information is configurable.

The **Delete After Selection** feature is used to automatically delete an order after it is selected from the Pending Orders list on the Patient ID screen. Disabling this feature will allow an order to be selected multiple times from the Patient ID screen.

To configure the Order settings:

1 Press down and hold the *Alt* key ( ) on the keyboard.

2 Press the *Q* key. The Orders screen appears.

3 Under **Quick Search Fields**, the **Patient ID** field is selected as the searchable field. To select another field as the searchable field, press the down arrow key ( ) until the field is highlighted.

NOTE Ensure that the field enabled as the Quick Search Field is also enabled as an active Patient ID field on the cardiograph (see page 2-12).

4 Press the *Tab* key ( ) or turn the Trim Knob until the **Delete After Selection** text is highlighted.

5 Press the space bar (on keyboard) or the Trim Knob to select or to clear the check box. With the check box selected, an order will be deleted from the Pending Orders list when it is selected during a patient session. With the check box cleared, the order may be selected multiple times from the Patient ID screen.

Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key ( ) to save the changes and to exit the Configuration screen. The new changes are applied.

Rhythm Settings

The Rhythm settings specify the default Rhythm report settings on the cardiograph. These settings include the size and the speed of the waveforms, and the individual leads that appear on the printed Rhythm report.

REPORT

Any of the default settings may be changed during an individual patient session by selecting the **Report** button on the Command Toolbar and manually changing the Rhythm report settings.

NOTE To set Rhythm filter settings, see the next section “Filter Settings.”

To set default Rhythm settings:

1. Press down and hold the *Alt* key ( ) on the keyboard.
2. Press the *V* key. The Rhythm screen appears.
3. Press the *Tab* key ( ) or turn the Trim Knob until the drop-down list next to **Speed (mm/s)** is highlighted (in blue).
4. Press the space bar (on keyboard) or the Trim Knob to open the drop-down list.
5. Press the up or down arrow key ( ) to scroll through the available settings.
6. Press the space bar or the Trim Knob to select a setting.
7. Repeat the procedure to set the **Size (mm/mv)** and **Size (V Leads)** settings.

NOTE The **Size (V Leads)** setting scales the size of the precordial leads to one-half or to the same (full) size as the limb leads.

8. Press the *Tab* key or turn the Trim Knob to highlight the **Select All** check box to select all twelve leads to print on the Rhythm report. Or, press the *Tab* key or turn the Trim Knob to highlight the **I** lead check box. Press the up or down arrow key or turn the Trim Knob to select individual leads from the list. Press the space bar or the Trim Knob to select an individual lead check box.

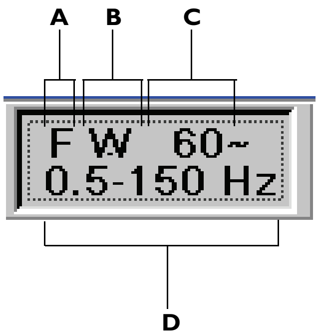
Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key ( ) to save the changes and to exit the Configuration screen. The new changes are applied.

Filter Settings

The filter settings specify the default filter settings that are used for each patient session. The configurable filter settings include:

- AC filter
- Artifact filter
- Baseline Wander filter
- High and Low Pass Frequency filters

Figure 2-9 The Filter Settings on the Status Bar



A Artifact Filter

B Baseline Wander Filter

C AC Filter

D High and Low Pass Frequency Filters

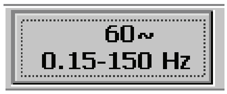
About the Artifact and Baseline Wander Filters

The Artifact and Baseline Wander filters are optional filters that may be enabled or disabled. These optional filters are enabled or disabled by selecting the **Filter** button on the Command Toolbar, see page 1-45.

The purpose of the Artifact filter is to remove skeletal muscle artifact from ECG signals. Skeletal muscle artifact is one of the most difficult noise sources to eliminate due to the fact that it contains the same frequencies as actual ECG signals. The Artifact filter will eliminate skeletal muscle artifact, but will also reduce all high frequency components of the ECG signal.

The Baseline Wander filter helps to suppress the slow drifting of the ECG signal during ECG recording. Baseline wander may result from patient respiration or from other sources.

Table 2-7 Artifact and Baseline Wander Filter Settings

Artifact and Baseline Wander Filters Enabled	Artifact and Baseline Wander Filters Disabled
	

About High and Low Pass Frequency Filter Settings

- The 0.5 Hz High Pass frequency filter setting provides the highest level of baseline wander suppression
- The 0.15 Hz High Pass frequency filter setting provides a moderate amount of baseline wander suppression (suitable for most ECGs) without distorting the ST segment
- The 0.05 Hz High Pass frequency filter setting provides the highest fidelity signal with the least amount of baseline wander suppression
- The 40 Hz Low Pass frequency filter setting provides the maximum amount of noise suppression at low fidelity
- The 100 Hz Low Pass frequency filter setting provides a moderate amount of noise suppression while providing accurate waveform representation
- The 150 Hz Low Pass frequency filter setting provides the highest fidelity with a minimal amount of high-frequency noise suppression

For more information on using filters, see Chapter 3 “The Patient Session” of the *PageWriter Trim Cardiograph Instructions for Use* on the User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com).

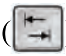
Factory Default Settings

- Artifact (**F**) Filter: On
- High Pass (Hz) Frequency: 0.15
- Low Pass (Hz) Frequency: 150
- AC Filter (**~**): On
- Rhythm Settings:
  - High Pass (Hz) Frequency: 0.15
  - Low Pass (Hz) Frequency: 150
  - Artifact (**F**) Filter: On

To configure filters:

1 Press and hold the *Alt* key () on the keyboard.

2 Press the *F* key. The Filter screen appears.

3 Press the *Tab* key () or turn the Trim Knob to highlight the **Filter On** setting.

4 Press the space bar (on keyboard) or the Trim Knob to select the check box (if not selected). Selecting this check box turns on the optional filters (Artifact, Baseline Wander or both) for all patient sessions.

5 The **Main Settings** section specifies the optional filters and high and low pass frequency filter settings used for all Auto ECGs. Press the *Tab* key or turn the Trim Knob to highlight the optional filters (Artifact, Wander, or both) that are turned on for all patient sessions.

6 Press the space bar or the Trim Knob to select the filter setting.

7 Press the *Tab* key or turn the Trim Knob to highlight a **High Pass** and **Low Pass** frequency setting that is used for all patient sessions.

TIP The High and Low Pass frequency settings of 0.15 and 150 Hz are recommended for most ECGs.

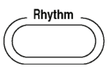
8 Press the space bar or the Trim Knob to select a setting.

9 Press the *Tab* key or turn the Trim Knob to highlight the **On** check box under **AC Filter (Hz)** (bottom of screen).

10 Press the space bar or the Trim Knob to select the **On** check box.

NOTE Clear the **On** check box to turn off the AC filter.

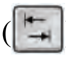
Setting Default Rhythm Filter Settings



Rhythm reports require a separate set of filter settings. The High and Low Pass frequency filters are automatically enabled for all Rhythm reports. The Artifact filter (**F**) is an optional filter that can be enabled during a patient session.

NOTE The 0.05 Hz High Pass Frequency setting is recommended for all Rhythm reports. This setting minimizes the potential for any ST segment distortion.

To set high and low pass frequency settings for rhythm reports:

1 Press the *Tab* key () or turn the Trim Knob to highlight a **High Pass (Hz)** and **Low Pass (Hz)** frequency filter setting.

2 Press the space bar or the Trim Knob to select the setting.

Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key () to save the changes and to exit the Configuration screen. The new changes are applied.

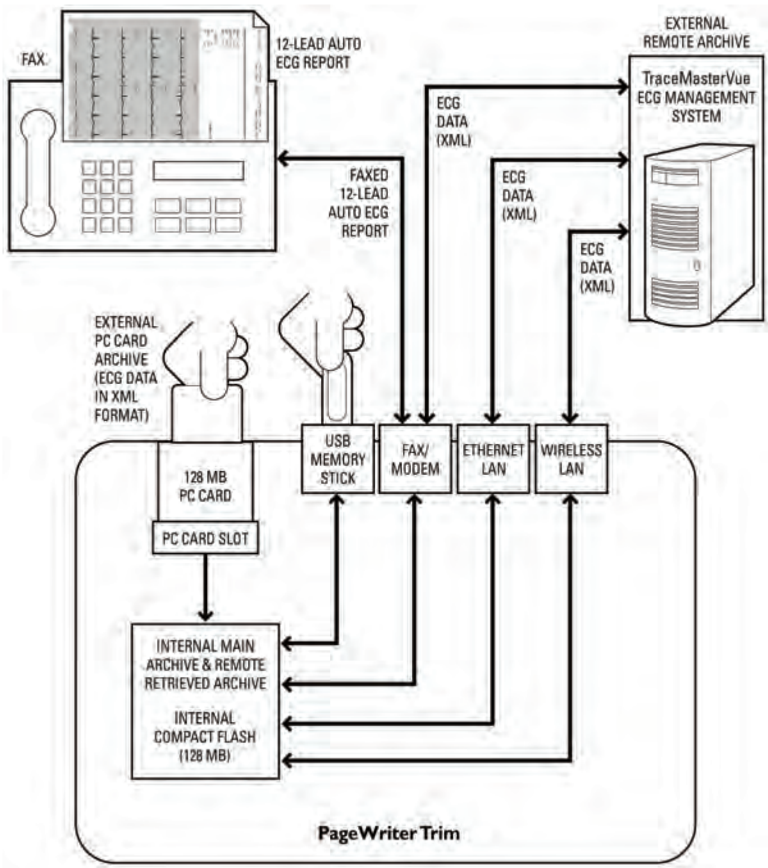
TraceMasterVue ECG Management System Remote Sites

The Remote Sites feature on the cardiograph is used with the Philips TraceMasterVue ECG Management System. The Remote Sites settings are used to configure how TraceMasterVue Remote Sites receive ECGs from the cardiograph, or transmit ECGs to the cardiograph. ECG transmission to and from a Remote Site can be done with a modem, or with a network connection.

The following illustration provides an overview of how Auto ECGs can be transferred from the cardiograph to a TraceMasterVue ECG Management System.

For detailed information on configuring the PageWriter Trim cardiograph with a TraceMasterVue ECG Management System, see the *Configuring Cardiograph and TraceMasterVue Communication Guide* available for download from the Philips InCenter web site (incenter.medical.philips.com). For information on using the Philips InCenter web site, see page 1-4

Figure 2-10 Transferring Auto ECGs from the cardiograph



TraceMasterVue Remote Sites that are defined in Configuration appear in the **Selected Archive** drop-down list in the Archive. The Archive is used to transmit ECGs to and from the cardiograph, and to search the Remote Site for ECGs to print on the cardiograph using the Query feature.

For more information on using the Archive, see Chapter 4 “PageWriter Trim II and III Orders and Archive” of the *PageWriter Trim Cardiograph*

Instructions for Use on the User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com).

Remote Site Fax and Modem Connection (USA and Canada only)

The cardiograph supports fax transmission of hard copy (non-editable) ECGs and modem transmission of ECG data to a TraceMasterVue ECG Management System. The cardiograph uses a PCMCIA Card modem (optional feature, see page 1-30) that supports printing to Group 3 Fax machines, and supports Class 1 or 2 Fax modem protocol (USA and Canada only).

For information on compatible PCMCIA card modems used outside of the USA and Canada, contact the nearest Philips Response Center (see page 1-50) or your local distributor.

WARNING **Never connect the PC card modem to a phone line when the cardiograph is connected to a patient.**

To configure a remote site with fax transmission:

1 Insert the PC card modem into the PC card slot on the rear of the cardiograph before configuring the fax or modem connection (see page 1-28)

1 Press and hold down the *Alt* key ( ) on the keyboard.

2 Press the *R* key. The Remote Sites screen appears.

3 Press the *Tab* key ( ) or turn the Trim Knob to highlight **Fax** or **Network** (modem, LAN).

4 Press the space bar (on keyboard) or the Trim Knob to select the setting.

For Fax:

1 Press the *Tab* key ( ) or turn the Trim Knob to highlight the **Name:** field. The cursor appears in the field.

2 Type in up to forty numbers or letters to represent the name of the sender of each fax transmission.

3 Press the *Tab* key or turn the Trim Knob to highlight the **Station ID** field. The cursor appears in the field.

4 Type in up to forty numbers or letters.

5 Press the *Tab* key or turn the Trim Knob to highlight the **Name** field. The cursor appears in the field.

6 Type in up to forty numbers or letters to represent the name of the recipient of each fax transmission.

7 Press the *Tab* key or turn the Trim Knob to highlight the **Fax Number** field. The cursor appears in the field.

8 Type in the receiving fax number.

9 Press the *Tab* key or turn the Trim Knob to highlight the drop-down list under **Dialing Devices**.

10 Press the space bar (on keyboard) or the Trim Knob to display the drop-down list.

11 Press the up or down arrow key () to highlight the installed modem card.

NOTE If the installed modem card does not appear on the drop-down list, ensure that the modem card is fully inserted into the PC card slot on the rear of the cardiograph. If the card is not installed, or was not fully inserted into the slot, press and hold down the On/Standby button for three seconds, then release it. The cardiograph shuts down. Push the On/Standby button again. After the R/T ECG screen appears, follow the procedure "Opening the Configuration Screen" on page 2-3 and then return to this procedure.

12 Press the space bar to select the modem card.

13 Press the *Tab* key or turn the Trim Knob to highlight the **Memo:** field. The cursor appears in the field.

14 Type in optional memo text to appear on the fax cover sheet.

A new Remote Site can be created using these settings, or any existing Remote Site may be saved with the new settings. Follow steps 15 to 18 to create a new Remote Site, or see the next procedure to edit an existing Remote Site.

15 At the top of the screen a check mark and the word **Modified** appears. Press the *Tab* key or turn the Trim Knob to highlight the **Add New** button.

16 Press the space bar or the Trim Knob. The **Add New Remote Site** window appears.

17 Type in a name for the Remote Site configured with these fax settings.

18 Press the *Tab* key or turn the Trim Knob to highlight the **OK** button.

19 Press the space bar or the Trim Knob. The new Remote Site with fax transmission is created.

Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key ( ) to save the changes and to exit the Configuration screen. The new changes are applied.

To change remote site fax settings:

1 Press the *Tab* key ( ) or turn the Trim Knob until the **Remote Sites Archive** drop-down list is highlighted.

2 Press the space bar (on keyboard) or the Trim Knob to display the drop-down list.

3 Press the up or down arrow key ( ) to scroll through the drop-down list.

4 Press the space bar or the Trim Knob to select a Remote Site.

5 Make all changes to the selected Remote Site on the screen.

6 When all changes are complete, a check mark and the word **Modified** appears at the top of the screen.

7 Press the *Tab* key or turn the Trim Knob until the **Save** button is highlighted.

8 Press the space bar or the Trim Knob to save the new settings to the selected Remote Site.

Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key ( ) to save the changes and to exit the Configuration screen. The new changes are applied.

Remote Sites Network Connection

The cardiograph supports the following network connections for Remote Sites:

- 10/100-Base T Ethernet using a PCMCIA network LAN card (see page 1-29)
- wireless LAN card (see page 1-29)

For more information on configuring network settings, see your Network Administrator.

For information on configuring the wireless LAN card option with the PageWriter Trim Cardiograph, see the *PageWriter Trim Cardiograph Wireless LAN Installation Instructions* on the User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com).

To configure a remote site with a network connection:

1. Press and hold down the *Alt* key ( ) on the keyboard.
2. Press the *R* key. The Remote Sites screen appears.
3. Press the *Tab* key ( ) or turn the Trim Knob to highlight **Network** (modem, LAN).
4. Press the space bar (on keyboard) or the Trim Knob to select the setting.
5. Press the *Tab* key or turn the Trim Knob to highlight the settings **Receive (Query)** and **Copy/Transfer** (left side of screen).

NOTE Selecting the **Receive (Query)** check box enables the user to search the Remote Site (in the Archive) for ECG files to print on the cardiograph. Selecting the **Copy/Transfer** check box enables the user to transfer ECGs from the Archive to a TraceMasterVue ECG Management System.

6. Under the **Network** tab (right of screen), press the *Tab* key or turn the Trim Knob to highlight the **Server URL** field. The cursor appears in the field.

7 Type in the URL address of the Remote Site server. The Server URL is the TraceMasterVue file server URL. The Server URL must be entered in the following format: http://computername/EMSCOMM/

Computername is the TraceMaster file server computer name or IP address. Examples of valid URLs include: http://tracemaster/EMSCOMM/ or http://10.101.2.42/EMSCOMM/.

8 Press the *Tab* key or turn the Trim Knob to highlight the **User Name** field. The cursor appears in the field.

9 Type in the TraceMasterVue User Name.

10 Press the *Tab* key or turn the Trim Knob to highlight the **Password** field. The cursor appears in the field.

11 Type in the Password for the TraceMasterVue User Name.

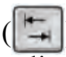
12 Press and hold down the *Alt* key on keyboard.

13 Press the *Z* key. The Net Connect screen appears.

14 Press the *Tab* key or turn the Trim Knob to highlight **Always Connected** or **Dial-up** (modem) connection.

15 Press the space bar or the Trim Knob the select the setting.

For Dial-up Modem:

1 Press the *Tab* key () or turn the Trim Knob to highlight the **Dialing Devices** drop-down list.

2 Press the space bar (on keyboard) or the Trim Knob to display the drop-down list.

3 Select the installed modem card.

NOTE If the installed modem card does not appear on the drop-down list, ensure that the modem card is fully inserted into the PC card slot on the rear of the cardiograph. If the card is not installed, or was not fully inserted into the slot, press and hold down the On/Standby button for three seconds, then release it. The cardiograph shuts down. Push the On/Standby button again. After the R/T ECG screen appears, follow the procedure “Opening the Configuration Screen” on page 2-3 and then return to this procedure.

4 Press the *Tab* key or turn the Trim Knob to highlight the **Phone** field. The cursor appears in the field.

5 Type in the dial-up modem telephone number.

6 Press the *Tab* key or turn the Trim Knob to highlight the **User Name** field. The cursor appears in the white box.

7 Type in the TraceMasterVue User Name for the dial-up modem connection.

8 Press the *Tab* key or turn the Trim Knob to highlight the **Password** field. The cursor appears in the field.

9 Type in the password for the TraceMasterVue user name.

10 Press the *Tab* key or turn the Trim Knob to highlight the **Domain** field. The cursor appears in the white box.

11 Type in the Domain name (optional).

12 At the top of the screen a check mark and the word **Modified** appears. Press the *Tab* key or turn the Trim Knob to highlight the **Add New** button.

13 Press the space bar or the Trim Knob. The **Add New Remote Site** window appears.

14 Type in a name for this Remote Site.

15 Press the *Tab* key or turn the Trim Knob to highlight the **OK** button.

16 Press the space bar or the Trim Knob.

The Remote Site appears in the **Remote Sites Archive** drop-down list (top of screen) and may be selected in the Archive.

Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key ( ) to save the changes and to exit the Configuration screen. The new changes are applied.

Network Settings

The settings in this section include network protocol settings used with the wireless LAN or the Ethernet LAN connection. The cardiograph supports the following network connections:

- 10/100-Base T Ethernet using a PCMCIA network card (see page 1-29)
- wireless LAN card (see page 1-29)

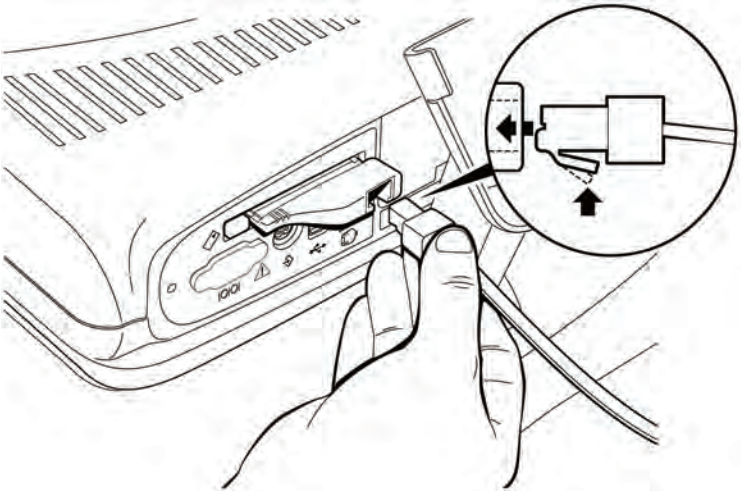
The network configuration settings support TCP/IP using a static IP address or a DNS or WINS server.

For more information on configuring network settings, see your Network Administrator.

For information on configuring the wireless LAN card option with the PageWriter Trim Cardiograph, see the *PageWriter Trim Cardiograph Wireless LAN Installation Instructions* on the User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com).

For detailed information on configuring the PageWriter Trim cardiograph with a TraceMasterVue ECG Management System, see the *Configuring Cardiograph and TraceMasterVue Communication Guide* available for download from the Philips InCenter web site (incenter.medical.philips.com). For information on using the Philips InCenter web site, see page 1-4

Figure 2-11 Connecting the LAN card



Networking Overview

The PageWriter Trim cardiograph communicates with the TraceMasterVue ECG Management System through a network connection. In the cardiograph network configuration settings, all TCP/IP settings are configurable based on the specific needs of the clinical environment. Both DHCP (Dynamic Host Configuration Protocol) and fixed IP (Internet Protocol) address settings are available.

DHCP

With DHCP, the LAN automatically provides a dynamic IP address to the cardiograph. The LAN logs the cardiograph's unique MAC (Media Access Control) address and provides a temporary DHCP IP address for the cardiograph.

If the cardiograph is connected to different LAN servers within the same facility, it may have to be reset in order to be recognized and assigned a DHCP IP address. Or, if the cardiograph has not been connected to the network for a specified period of time (defined by the LAN), the network may not recognize the cardiograph. In this case, the cardiograph may also have to be reset. Resetting the cardiograph at multiple locations on the LAN may be time consuming.

If the cardiograph is going to be connected to multiple sites within a facility, or to different LAN servers within a facility, it may be more efficient to assign the cardiograph a fixed IP address.

Fixed IP Address

Fixed IP addresses allow the cardiograph to be recognized throughout a network. With a fixed IP address, the cardiograph will be recognized at multiple locations throughout a LAN and will not require a full reset in order to be recognized. Using fixed IP addresses will also avoid the problem of having multiple IP addresses assigned to a single cardiograph.

Auto Negotiation

When the cardiograph is connected to a LAN, it automatically configures the correct settings for Ethernet speed and mode (half or full duplex). If the auto-negotiation fails, it may be necessary to lock a specific switch or router port to a fixed setting (for example, 100BaseT Full Duplex) in order to obtain a connection.

Issues with the timing of the PageWriter Trim auto negotiation have been reported with certain CISCO switches (such as the CISCO 4506). To obtain a LAN connection with these switches, the specific LAN port may have to be locked down with fixed Ethernet port settings. The cardiograph should only be connected to these locked down ports.

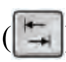
For more information on networking options or to troubleshoot specific networking issues, consult your network administrator.

To configure cardiograph network settings:

1 Press and hold down the *Alt* key (

) on the keyboard.

2 Press the *N* key. The Network screen appears.

3 Under **Network ID**, press the *Tab* key () or turn the Trim Knob to highlight the text field next to **Computer Name** (if not highlighted).

4 Type in the computer name for the cardiograph (up to sixteen letters or numbers).

NOTE The Computer Name may not contain spaces.

5 Press the *Tab* key or turn the Trim Knob to highlight the **Obtain IP Address Automatically** (DNS) or **Specify IP Address** (fixed IP address) setting depending on your network requirements.

6 Press the space bar or the Trim Knob to select the setting.

For Fixed IP address:

1 Press the *Tab* key ( ) or turn the Trim Knob to select the **IP Address**, **Subnet Mask**, and **Default Gateway** fields. A cursor appears in the white box. Type in the IP address information for each field.

2 Press the *Tab* key or turn the Trim Knob to select the **Primary DNS Server** or **Primary WINS** fields. The cursor appears in the white box.

3 Type in the server information.

4 Press the *Tab* key or turn the Trim Knob to select the **Secondary DNS** or **Secondary WINS** fields (if necessary).

5 Type in the server information.

6 To apply the new network settings the cardiograph must restart. Press the *Enter* key ( ) on keyboard to save the settings.

7 Press and hold the On/Standby button for three seconds and then release the button. The message **PageWriter Trim is shutting down...** appears. The cardiograph shuts off.

8 Press the On/Standby button to restart the cardiograph. The cardiograph restarts and the new network settings are applied.

Password Settings

The following cardiograph features can be configured to be password access controlled:

- Archive
- Configuration

Passwords can be up to six letters or numbers in length and are case-sensitive (capital and lower case letters can be used). The **Password Configuration System** password must be configured first. This password controls access to all password settings on the Configuration screens. Entering a correct password allows the user to change all password settings on the cardiograph. The **Archive** password must be entered to open the Archive screen. The **Configuration** password must be entered to open the Configuration screens.

To configure passwords:

1 Press and hold down the *Alt* key ( ) on keyboard.

2 Press the *W* key. The Passwords screen appears. The **Password Configuration System** password needs to be configured first.

3 Press the *Tab* key ( ) or turn the Trim Knob until the first field to the right of **Password Configuration System** (under the **Enter Password** column) is highlighted. The cursor appears in the field.

4 Type in a password (up to six letters or numbers in length).

5 Press the *Tab* key or turn the Trim Knob until the cursor appears in the second field (under the **Confirm Password** column).

6 Re-enter the password.

NOTE If the password is re-entered incorrectly, the message **Invalid Password Entered** appears. Push the Trim Knob to select **OK**. Re-enter the password.

7 Press the *Tab* key or turn the Trim Knob to highlight the **On** check box to the right of **Password Configuration System** (under the **Enabled** column).

8 Follow the same procedure to configure passwords for the Archive and Configuration features on the cardiograph.

Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key ( ) to save the changes and to exit the Configuration screen. The new changes are applied.

Lost Passwords

If a cardiograph password is lost and cannot be retrieved, contact the nearest Philips Response Center for assistance, see page 1-50.

Locale Settings

The settings in this section include setting time format, date format, numerical, and unit measurements for specific countries or regions.

To change the date and time:

1. Press the hold down the *Alt* key (Alt) on the keyboard.
2. Press the *L* key. The Locale screen appears.
3. The **Region:** drop-down list is highlighted. Press the space bar (on keyboard) or the Trim Knob to display the drop-down list.
4. Press the up or down arrow key (↑ ↓) to scroll through the language and country selections.
5. Press the space bar to select a language and country setting. The settings for the specified language and country appear on the screen.
6. Any of the settings on the screen may be edited, if necessary.

   Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key (↵) to save the changes and exit the Configuration screen. The new changes are applied.

Cardiograph Settings

The optional settings in this section provide information about a specific cardiograph including: **Location Code** (LOC), **Device ID** (identification number to track a specific cardiograph), **Institution Label**, **Institution Number**, **Facility Label**, **Facility Number**, **Department Label** and **Department Number**.

Each of these fields may be customized with any thirty-two letters or numbers. **Location Code** (LOC) is limited to five numbers only.

Each of these configured information fields appear on the printed Auto ECG and are transferred with the ECG to a TraceMasterVue ECG Management System. To view examples of these information fields on the printed ECG report see Chapter 5 “Reading the Printed ECG Report” of the *PageWriter Trim Cardiograph Instructions for Use* on the User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com). The table below describes each cardiograph identification information field.

Table 2-8 Cardiograph Identification Information Fields

Cardiograph Identification Field	Description
LOC (Location Code)	▪ Only used with a TraceMaster ECG Management System (version A.04.02 or higher) ▪ Limited to five numbers in length ▪ Does not appear on the printed ECG ▪ Configuring this information field disables the **Institution Number** and **Department Number** fields
Device ID	▪ Appears on the printed ECG ▪ Limited to thirty-two letters or numbers in length
Institution Label	▪ Appears on the printed ECG ▪ Limited to thirty-two letters or numbers in length
Institution Number	▪ Appears on the printed ECG ▪ Limited to thirty-two numbers or in length
Facility Label	▪ Appears on the printed ECG ▪ Limited to thirty-two letters or numbers in length
Facility Number	▪ Appears on the printed ECG ▪ Limited to thirty-two numbers in length

Table 2-8 Cardiograph Identification Information Fields *(continued)*

Cardiograph Identification Field	Description
Department Label	■ Appears on the printed ECG ■ Limited to thirty-two letters or numbers in length
Department Number	■ Appears on the printed ECG ■ Limited to thirty-two numbers or length

Factory Default Setting

- No cardiograph information fields are entered

To configure institution information

1 Under **Identification**, press the *Tab* key ( ) or turn the Trim Knob until the field next to **Location Code** is selected. The cursor appears in the field.

NOTE The **Location Code (LOC)** is only used with the Philips TraceMaster ECG Management System (version A.04.02 or higher) and does not appear on the printed 12-lead ECG. Configuring the LOC disables the **Institution Number** and **Department Number** fields. For more information see the *Configuring TraceMasterVue and Cardiograph Communication Guide* on the User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com).

2 Enter the Location Code.

NOTE The **Location Code** is limited to five numbers in length. The first three numbers are used to identify an institution number, and the last two numbers are used to identify a department number.

3. Press the *Tab* key or turn the Trim Knob to select the field next to **Device ID**. The cursor appears in the field.
4. Enter the number.
5. Press the *Tab* key or turn the Trim Knob to select the field next to **Institution Label**. The cursor appears in the field.
6. Enter the text.
7. Press the *Tab* key or turn the Trim Knob to select the field next to **Institution Number**. The cursor appears in the field.
8. Enter the number.
9. Press the *Tab* key or turn the Trim Knob to select the field next to **Facility Label**. The cursor appears in the field.
10. Enter the text.
11. Press the *Tab* key or turn the Trim Knob to select the field next to **Facility Number**. The cursor appears in the field.
12. Enter the number.
13. Press the *Tab* key or turn the Trim Knob to select the field next to **Department Label**. The cursor appears in the field.
14. Enter the text.
15. Press the *Tab* key or turn the Trim Knob to select the field next to **Department Number**. The cursor appears in the field.
16. Enter the number.

Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key () to save the changes and to exit the Configuration screen. The new changes are applied.

Metronome Settings

This section is also used to configure the **Metronome** feature. The Metronome is used with the Auto ECG function to perform exercise stress tests such as the Masters' Two Step. Using this feature, the patient exercises at a rate indicated by the metronome chime, and an Auto ECG is taken after the patient stops exercising. The default settings for the Metronome are set in Configuration, but these settings may be changed by the user during a patient session.

For more information on using the Metronome feature during a patient session, see Chapter 3 "The Patient Session" of the *PageWriter Trim Cardiograph Instructions for Use* on the User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com).

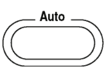
Factory Default Settings

Metronome

- Set to Off
- Number of Steps 50
- Stress Test Protocol: Master Double

To set metronome settings:

1 Press the *Tab* key () or turn the Trim Knob until the **ON** check box under **Metronome** is highlighted.



2 Press the space bar (on keyboard) or the Trim Knob to select the check box. Selecting the **ON** check box enables the Metronome feature. Each time the **Auto** button is pressed, the Metronome window appears on the screen.

3 Press the *Tab* key or turn the Trim Knob to highlight the field to the right of **Number of Steps** (under **Speed**). Type in the number of default steps (16-70) to be used with the Metronome feature during a typical stress test. This setting can be changed by the user during a patient session.

4 Press the *Tab* key or turn the Trim Knob to highlight a setting under **Stress Test Protocol.** This setting sets the default Masters' Two Step stress test used with the Metronome. This setting can be changed by the user during a patient session.

Table 2-9 Masters' Two Step Stress Test Settings

Test	Description
Master Single	■ Patient exercises for 90 seconds
Master Double	■ Patient exercises for 180 seconds

5 Press the *Tab* key or the Trim Knob to select a Masters' Two Step Stress Test setting.

Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key () to save the changes and to exit the Configuration screen. The new changes are applied.

Magnetic Card Reader Settings

This section also sets the default Patient ID field settings for the optional magnetic card reader, see page 1-33 for more information.

TIP Ensure that the selected Patient ID fields (see page 2-12) and the Patient ID fields selected for the magnetic card reader are consistent.

The magnetic card reader supports the following standards:

- ISO 7810, 7811-1, -2, -3, -4, -5
- JIS X 6301 and X 6302

Factory Default Settings

- Magnetic Card Reader:
  - Field 1: Patient ID (number)
  - Fields 2, 3, 4: no fields selected (None).

To configure the magnetic card reader:

1. Press the *Tab* key ( ) or turn the Trim Knob until the **Field 1** drop-down list under **Magnetic Card Reader - Track 1** is highlighted.
2. Press the space bar (on keyboard) or the Trim Knob to display the drop-down list of available Patient ID information fields.
3. Press the up or down arrow key ( ) to scroll through the drop-down list.
4. Press the space bar or the Trim Knob to select a Patient ID field.
5. Repeat the procedure for Fields 2-4.

   Select another tab (top of screen) to continue making changes. If done making changes, press the *Enter* key ( ) to save the changes and to exit the Configuration screen. The new changes are applied.

Saving Configured Settings

All cardiograph configured settings may be saved as a *configuration settings file* to a PC card or to a USB memory stick, and then uploaded to multiple cardiographs. Use this feature when configuring multiple cardiographs with the same settings.

Select all of the default cardiograph configuration settings before saving them to a PC card or to a USB memory stick as a configuration file.

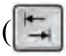
NOTE Network settings are not saved with the configuration settings file. It is recommended to print out a configuration settings report to capture a record of all network settings on the cardiograph, see page 2-68.

TIP It is recommended to save cardiograph configuration settings to a PC card or to a USB memory stick in case of system failure.

To save configuration settings to a PC card:

1 Select all cardiograph configured settings to be saved to the configuration settings file.

2 Insert the PC card into the PC card slot, or insert the USB memory stick into the USB port on the rear of the cardiograph.

3 Press the *Tab* key () or turn the Trim Knob to highlight the **Config** button on the Command Toolbar.

4 Press the space bar (on keyboard) or the Trim Knob to select the button. The Configuration screen appears.

5 Press and hold down the *Alt* key () on the keyboard.

6 Press the *S* key. The System screen appears.

7 Press the *Tab* key or turn the Trim Knob until the **Save Configuration File** button under **Configuration Management** is highlighted.

8 Press the space bar or the Trim Knob. The **Save Configuration File** window appears.

9 The **Select Destination Media** drop-down list is highlighted. Press the up or down arrow key to select the output media.

10 Press the *Tab* key or turn the Trim Knob to select the **Input file name**: field.

11 Type in a name for the configuration settings file.

12 Press the *Tab* key or turn the Trim Knob until the **OK** button is highlighted.

13 Press the space bar or the Trim Knob to select the button.

A message appears that the configuration settings file is saved to the selected media. The Configuration screen automatically closes. The configuration settings file may be uploaded to additional cardiographs using the following procedure.

Loading the Configuration Settings File

If the cardiograph loses all configured settings, upload the cardiograph configuration settings file to the cardiograph using a PC card or USB memory stick.

This procedure may also be used when loading cardiograph configuration settings onto multiple cardiographs.

NOTE Network settings are not saved with the configuration settings file. It is recommended to print out a configuration settings report to capture a record of all network settings on the cardiograph, see page 2-68.

To load the configuration settings file onto the cardiograph:

1. Insert the PC card or the USB memory stick that contains the configuration settings file into the PC card slot or the USB port on the rear of the cardiograph.

2. From the R/T ECG screen, press the *Tab* key ( ) or turn the Trim Knob to select the **Config** button on the Command Toolbar.
3. Press the space bar (on keyboard) or the Trim Knob to select the button. The Configuration screen appears.
4. Press and hold down the *Alt* key ( ) on the keyboard.
5. Press the *S* key. The System screen appears.
6. Press the *Tab* key or turn the Trim Knob until the **Load Configuration File** button under **Configuration Management** is highlighted.
7. Press the space bar or the Trim Knob. The **Load Configuration File** window appears. The **Select Input Source** pull-down list is highlighted.
8. Press the up or down arrow key ( ) to select the input media. Press the space bar to select the input media. The configuration settings file(s) saved to the input media display under **Profiles List**.

9 Press the *Tab* key or turn the Trim Knob to highlight a profile under **Profiles List**. Press the space bar or the Trim Knob to select the configuration settings file.

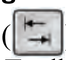
10 Press the *Tab* key or turn the Trim Knob to highlight the **OK** button. Press the space bar or the Trim Knob to select the button.

The configuration settings file is loaded onto the cardiograph. The new settings are applied.

Restoring Factory Default Configuration Settings

The factory default settings on the cardiograph can be restored at any time. Ensure that before reverting to factory default settings that any custom configuration settings are saved to a PC card or to a USB memory stick (see page 2-64).

To restore factory default configuration settings:

1 From the R/T ECG screen, press the *Tab* key () or turn the Trim Knob to select the **Config** button on the Command Toolbar.

2 Press the space bar (on keyboard) or the Trim Knob to select the button. The Configuration screen appears.

3 Press and hold down the *Alt* key () on the keyboard.

4 Press the *S* key. The System screen appears.

5 Press the *Tab* key or turn the Trim Knob until the **Reset to Default** button under **Configuration Management** is highlighted.

6 Press the space bar or the Trim Knob. The **Configuration Management** window appears with the message **Are you sure you want to reset to factory settings?** Press the space bar or the Trim Knob to select the **Yes** button.

NOTE Select the **No** button to not apply the factory default settings and to exit the window.

7 The message **System reset to factory default settings** appears. The factory default configuration settings are loaded onto the cardiograph.

Printing the Configuration Settings Report

A report that lists all of the configured settings information for the cardiograph may be printed from the Configuration screen. This report does include all TraceMasterVue Remote Site and network configuration settings.

To print a configuration settings report:

1 From the R/T ECG screen, press the *Tab* key ( ) or turn the Trim Knob to select the **Config** button on the Command Toolbar.

2 Press the space bar (on keyboard) or the Trim Knob to select the button. The Configuration screen appears.

3 Press and hold down the *Alt* key ( ) on the keyboard.

4 Press the *S* key. The System screen appears.

5 Press the *Tab* key or turn the Trim Knob until the **Print Configuration File** button under **Configuration Management** is highlighted. Press the space bar or the Trim Knob to select the button.

6 The **Configuration Print Setup** window appears. Press the *Tab* key or turn the Trim Knob until the **Select Configuration Item** drop-down list is highlighted. Press the space bar to open the drop-down list.

7 Press the up or down arrow key ( ) to select an individual item to print on the report, or specify **Select All** to print out all configured settings information. Press the space bar to select the printed report settings and to close the drop-down list.

8 Press the *Tab* key or the Trim Knob to highlight the **OK** button. Press the space bar or the Trim Knob to select the button. The configuration settings report prints.

A

Suppressed Borderline Interpretive Statements

Introduction

This Appendix includes a listing of all borderline interpretive statements that are suppressed using the *Borderline Statement Suppression* feature that is available with the Philips 12-Lead Algorithm version PH090A. This feature is not available with the algorithm version number PH080A.

For more information on the Philips 12-Lead Algorithm, including a listing of all interpretive, reason, and severity statements included in the Philips 12-Lead Algorithm, please see the *Philips 12-Lead Algorithm Physician's Guide* on the User Documentation CD, or download the file from the Philips InCenter web site (incenter.medical.philips.com). For information on using the Philips InCenter web site, see page 1-4.

Level 1 Suppressed Statements

The following interpretive statements listed in Table A-1 are suppressed when the **Level - 1 few** setting is selected under the **Analysis** tab on the Configuration screen. For information on configuring the Borderline Statement Suppression feature, see "About the Borderline Statement Suppression Setting" on page 2-10.

Table A-1 Level 1 Setting - Suppressed Statements

Statement Code	Interpretive Statement
BAVCD	BORDERLINE AV CONDUCTION DELAY
BIVCD	BORDERLINE INTRAVENTRICULAR CONDUCTION DELAY
CLAA	CONSIDER LEFT ATRIAL ABNORMALITY
CRAA	CONSIDER RIGHT ATRIAL ABNORMALITY
ET	EARLY PRECORDIAL R/S TRANSITION
ETRSR1	RSR' IN V1 OR V2, RIGHT VCD OR RVH
LOWT	BORDERLINE T WAVE ABNORMALITIES
LT	LATE PRECORDIAL R/S TRANSITION
LVHQ	CONSIDER LEFT VENTRICULAR HYPERTROPHY
LVOLFB	BORDERLINE LOW VOLTAGE IN FRONTAL LEADS
NFAD	NO FURTHER ANALYSIS ATTEMPTED DUE TO PACED RHYTHM
NFRA	NO FURTHER RHYTHM ANALYSIS ATTEMPTED DUE TO PACED RHYTHM
QMAB	ARTIFACT IN LEAD(S) AND BASELINE WANDER IN LEAD(S)
QMART	ARTIFACT IN LEAD(S)
QMBW	BASELINE WANDER IN LEAD(S)
REPB	BORDERLINE REPOLARIZATION ABNORMALITY

Table A-1 Level 1 Setting - Suppressed Statements *(continued)*

Statement Code	Interpretive Statement
RSR1	RSR' IN V1 OR V2, PROBABLY NORMAL VARIANT
RSRNV	RSR' IN V1, NORMAL VARIATION
SDALP	NONSPECIFIC ST DEPRESSION, ANTEROLATERAL LDS
SDANP	NONSPECIFIC ST DEPRESSION, ANTERIOR LEADS
SDCU	MINIMAL ST DEPRESSION
SDINP	NONSPECIFIC ST DEPRESSION, INFERIOR LEADS
SDJ	JUNCTIONAL ST DEPRESSION
SDM	MINIMAL ST DEPRESSION
SEALP	T ELEVATION, PROB NORMAL VARIATION, ANT-LAT
SEANP	ST ELEV, PROBABLY NORMAL VARIATION, ANT LEADS
SEINP	T ELEVATION, PROBABLY NORMAL VARIATION, INF
SPRB	BORDERLINE SHORT PR INTERVAL
TAXAB	BORDERLINE T WAVE ABNORMALITIES
TAXQT	BORDERLINE T WAVE ABNORMALITIES
TTW1	TALL T, PROBABLY NORMAL VARIANT, ANT-LAT LDS

Level 2 Suppressed Statements

The interpretive statements listed in Table A-1, "Level 1 Setting - Suppressed Statements," on page A-2, and the interpretive statements listed in Table A-2 beginning on this page, are suppressed when the **Level - 2 many** setting is selected under the **Analysis** tab on the Configuration screen. For information on configuring the Borderline Statement Suppression feature, see "About the Borderline Statement Suppression Setting" on page 2-10.

Table A-2 Level 2 Setting - Suppressed Statements

Statement Code	Interpretive Statement
AMI1	BORDERLINE R WAVE PROGRESSION, ANTERIOR LEADS
AMI3	Q WAVE IN V1
AXL	BORDERLINE LEFT AXIS DEVIATION
AXR	BORDERLINE RIGHT AXIS DEVIATION
BIVCDL	BORDERLINE IVCD WITH LAD
CRHPI	CONSIDER RVH OR POSTERIOR INFARCT
CRHPIR	CONSIDER RVH OR PMI W/ SEC REPOL ABNORMALITY
CRPMI	TALL R WAVE IN V2, CONSIDER RVH OR PMI
CRVH	CONSIDER RIGHT VENTRICULAR HYPERTROPHY
IMI3	BORDERLINE INFERIOR Q WAVES
IMI4	CONSIDER LAFB OR INFERIOR INFARCT
IMI18	INFERIOR Q WAVES, PROBABLY NORMAL VARIATION
IRBBRV	IRBBB, THE RSR' PATTERN MAY ALSO REFLECT RVH

Table A-2 Level 2 Setting - Suppressed Statements *(continued)*

Statement Code	Interpretive Statement
LMI10	BORDERLINE LATERAL Q WAVES
LMI49	LATERAL Q WAVES, PROBABLY NORMAL VARIATION
LQTB	BORDERLINE PROLONGED QT INTERVAL
LVHR56	LVH BY VOLTAGE
LVHR6	LVH BY VOLTAGE
LVHRS	CONSIDER LEFT VENTRICULAR HYPERTROPHY
LVHRSI	LVH BY VOLTAGE
LVHS12	LVH BY VOLTAGE
LVHTA	CONSIDER LEFT VENTRICULAR HYPERTROPHY
LVHV	LVH BY VOLTAGE
MSTEA	MINIMAL ST ELEVATION, ANTERIOR LEADS
MSTEAL	MINIMAL ST ELEVATION, ANTEROLATERAL LEADS
MSTEI	MINIMAL ST ELEVATION, INFERIOR LEADS
MSTEL	MINIMAL ST ELEVATION, LATERAL LEADS
PLAA	PROBABLE LEFT ATRIAL ABNORMALITY
PQAL	BORDERLINE Q WAVE IN ANTEROLATERAL LEADS
PQAN	BORDERLINE Q WAVE IN ANTERIOR LEADS
PQIN	BORDERLINE Q WAVES IN INFERIOR LEADS

Table A-2 Level 2 Setting - Suppressed Statements *(continued)*

Statement Code	Interpretive Statement
PQLA	BORDERLINE Q WAVES IN LATERAL LEADS
PRAA	PROBABLE RIGHT ATRIAL ABNORMALITY
REPBAL	BORDERLINE REPOL ABNORMALITY, ANT-LAT LEADS
REPBAN	BORDERLINE REPOL ABNORMALITY, ANT LEADS
REPBIL	BORDERLINE REPOL ABNORMALITY, INF-LAT LEADS
REPBIN	BORDERLINE REPOL ABNORMALITY, INFERIOR LEADS
REPBLA	BORDERLINE REPOL ABNORMALITY, LATERAL LEADS
RVHS5	CONSIDER RIGHT VENTRICULAR HYPERTROPHY
RVHS6	CONSIDER RIGHT VENTRICULAR HYPERTROPHY
SD0AL	MINIMAL ST DEPRESSION, ANTEROLATERAL LEADS
SD0AN	MINIMAL ST DEPRESSION, ANTERIOR LEADS
SD0IN	MINIMAL ST DEPRESSION, INFERIOR LEADS
SD0LA	MINIMAL ST DEPRESSION, LATERAL LEADS
SD0NS	MINIMAL ST DEPRESSION
SPR	SHORT PR INTERVAL, ACCELERATED AV CONDUCTION

Table A-2 Level 2 Setting - Suppressed Statements *(continued)*

Statement Code	Interpretive Statement
SQT	SHORT QT INTERVAL
T0AL	BORDERLINE T ABNORMALITIES, ANT-LAT LEADS
T0AN	BORDERLINE T ABNORMALITIES, ANTERIOR LEADS
T0IN	BORDERLINE T ABNORMALITIES, INFERIOR LEADS
T0LA	BORDERLINE T ABNORMALITIES, LATERAL LEADS
T0NS	BORDERLINE T WAVE ABNORMALITIES
TTW30	TALL T WAVES, PROBABLY NORMAL VARIANT

Index

O

P

Q

R

S

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